

CSC 2032 – Object-Oriented Programming

PROJECT REPORT

BOARDING FINDER SYSTEM

A Web-Based Boarding Room Management System

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Finding suitable boarding accommodation is one of the major challenges faced by university students, especially those who study away from their hometowns. In many cases, students rely on social media, notice boards, friends, or personal visits to find available boarding rooms. These traditional methods are often time-consuming, inefficient, and may not provide accurate or updated information regarding room availability, rental prices, or facilities.

Similarly, boarding house owners experience difficulties in advertising their available rooms and managing booking requests. They usually depend on word-of-mouth recommendations or printed advertisements, which limit their ability to reach a wider group of potential tenants. Furthermore, manually managing room availability and booking records increases the possibility of errors and double bookings.

With the rapid growth of information technology and internet accessibility, web-based applications have become an effective solution for managing accommodation services. Such systems enable users to access information anytime and from anywhere while providing real-time updates. A centralized online platform can simplify communication between students and boarding owners, reduce paperwork, and improve the overall efficiency of the boarding room reservation process.

The **Boarding Finder** system was developed as a web-based application to address these issues. The system allows students to search for boarding rooms using various criteria such as university, gender preference, location, and monthly rental fee. Students can also view detailed room information, submit booking requests, and monitor their booking status through a personal dashboard.

On the other hand, boarding owners can register their properties, upload room images, manage room details, review booking requests, and monitor room occupancy through a dedicated owner dashboard. The system automatically updates room availability based

on the occupancy level, reducing the possibility of overbooking and ensuring accurate booking management.

The application was developed using **Java Server Pages (JSP)**, **Java Servlets**, **MySQL**, **Bootstrap**, and **Apache Tomcat**, following an object-oriented programming approach. These technologies provide a reliable, secure, and scalable environment for developing modern web applications.

Overall, the Boarding Finder system offers a convenient, efficient, and user-friendly solution that benefits both students and boarding house owners while improving the management of boarding accommodation services.

1.2 Problem Statement

Finding boarding accommodation is a significant challenge for many university students. Most students rely on traditional methods such as asking friends, searching through social media, or visiting boarding houses in person. These methods require considerable time and effort and often fail to provide accurate or up-to-date information regarding room availability and facilities.

Boarding owners also face several challenges in managing their properties. Manual record keeping, telephone inquiries, and face-to-face communication make it difficult to organize booking requests efficiently. As a result, owners may experience duplicate bookings, inaccurate occupancy records, and communication delays.

Another limitation of the traditional boarding management process is the lack of a centralized platform where students and owners can interact. Students cannot easily compare different boarding options, while owners have limited opportunities to advertise their available rooms to a wider audience.

To overcome these challenges, there is a need for an online boarding room management system that enables students to search and reserve boarding rooms conveniently while allowing owners to manage room information and bookings efficiently. Such a system should provide accurate room availability, simplify the booking process, and improve communication between students and boarding owners.

1.3 Aim of the Project

The main aim of this project is to develop a web-based boarding room management system that enables university students to search, view, and reserve boarding rooms online while allowing boarding owners to manage room listings, booking requests, and occupancy efficiently.

1.4 Objectives of the Study

The primary objectives of the Boarding Finder system are:

General Objective

To develop a web-based boarding room management system that simplifies the boarding room reservation process for students and improves room management for boarding owners.

Specific Objectives

- To provide an online platform for searching available boarding rooms.
- To allow students to filter rooms based on university, gender preference, location, and monthly rental fee.
- To enable students to register, log in, and submit booking requests.
- To provide detailed information about boarding rooms, including facilities, images, rental price, and location.
- To allow boarding owners to add, edit, and delete room information.
- To enable owners to accept or reject booking requests.
- To automatically manage room occupancy and availability.
- To reduce duplicate bookings through automated booking management.
- To provide separate dashboards for students and boarding owners.
- To improve the overall efficiency of boarding room management.

1.5 Scope of the Project

The Boarding Finder system is designed for university students and boarding house owners in Sri Lanka. The system operates as a web-based application that can be accessed using a standard web browser with an internet connection.

Students can create accounts, search for boarding rooms, view room details, submit booking requests, and manage their bookings through the student dashboard. Boarding owners can register their properties, manage room information, upload room images, review booking requests, and monitor room occupancy through the owner dashboard.

The system includes user authentication, room management, booking management, image uploading, Google Maps location links, occupancy management, and dark mode support. However, online payment processing and real-time messaging between students and owners are not included in the current version of the system.

1.6 Significance of the Study

The Boarding Finder system provides several benefits for both students and boarding house owners.

For students, the system reduces the time required to find suitable accommodation by providing an online platform with detailed room information and search filters. Students can compare different boarding options, submit booking requests, and monitor their booking status without visiting multiple boarding houses.

For boarding owners, the system provides an efficient method for advertising available rooms and managing bookings electronically. Automatic occupancy tracking reduces manual errors and prevents overbooking.

From an academic perspective, the project demonstrates the practical application of web technologies, object-oriented programming principles, and database management techniques. It also contributes to improving digital solutions for accommodation management within educational institutions.

1.7 Limitations of the Study

Although the Boarding Finder system provides a practical solution for boarding room management, several limitations exist.

- The system requires an internet connection to access its features.
- Online payment functionality is not available.
- Direct messaging between students and boarding owners is not implemented.
- Mobile application support is not included.
- Email and SMS notifications are not currently available.
- The system depends on users providing accurate room information during registration and updates.

1.8 Organization of the Report

This report is organized into six chapters.

- **Chapter One** introduces the project by presenting the background, problem statement, objectives, scope, significance, and limitations of the study.
- **Chapter Two** reviews previous studies and discusses the technologies used in developing the system.
- **Chapter Three** explains the system analysis and design, including the software architecture, database design, UML diagrams, and system requirements.
- **Chapter Four** describes the implementation of the Boarding Finder system and explains each module developed.
- **Chapter Five** presents system testing, test cases, results, and screenshots of the implemented system.
- **Chapter Six** concludes the report by summarizing the project achievements, discussing limitations, and providing recommendations for future improvements.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

A literature review is an essential part of any software development project because it provides an understanding of existing systems, related technologies, and previous research relevant to the proposed solution. It helps identify the strengths and weaknesses of existing methods while justifying the need for the new system.

The Boarding Finder system was developed to improve the process of searching and managing boarding accommodations for university students. The system integrates modern web technologies such as Java Server Pages (JSP), Java Servlets, MySQL, Bootstrap, and Apache Tomcat to provide a reliable and user-friendly platform. This chapter discusses previous approaches to boarding room management and the technologies used to develop the proposed system.

2.2 Existing Boarding Room Management Systems

Traditionally, boarding room management has been handled manually. Students usually search for boarding houses through newspaper advertisements, social media posts, university notice boards, or recommendations from friends and relatives. After identifying a potential boarding house, students must contact the owner personally or visit the location to inspect the room and confirm its availability.

Although this approach has been widely used for many years, it presents several challenges. Students often spend a significant amount of time searching for suitable accommodation, and the information they receive may be outdated or incomplete. In many cases, the advertised room may no longer be available when the student contacts the owner.

Boarding owners also encounter difficulties in managing room availability and booking requests. Maintaining handwritten records or spreadsheets increases the possibility of errors and makes it difficult to monitor occupancy accurately. Communication through

telephone calls and text messages further complicates the booking process, particularly when multiple students request the same room.

Several online accommodation platforms exist; however, many are designed for hotels or short-term rentals and do not specifically address the requirements of university boarding houses. These systems may include unnecessary features while lacking functionalities such as gender-based room allocation, university-based searching, or occupancy management.

Therefore, there is a need for a dedicated boarding room management system that focuses specifically on university students and boarding owners.

2.3 Web-Based Information Systems

A web-based information system is an application that operates through a web browser using internet technologies. Unlike traditional desktop applications, web applications do not require installation on individual computers. Users can access the system from any location using devices such as desktop computers, laptops, tablets, or smartphones.

Web-based systems offer several advantages, including centralized data storage, easier maintenance, platform independence, and real-time information updates. Since all information is stored within a central database, users always access the latest available data.

The Boarding Finder system is implemented as a web application because it enables students to search for boarding rooms anytime and anywhere. Similarly, boarding owners can manage room information and booking requests without needing specialized software.

2.4 Java Programming Language

Java is one of the most popular object-oriented programming languages used for developing desktop, web, and enterprise applications. It was developed by Sun Microsystems and is currently maintained by Oracle Corporation.

One of Java's main advantages is platform independence through the Java Virtual Machine (JVM), allowing programs to run on different operating systems without significant modifications.

Java provides several features that make it suitable for large-scale software development, including:

- Object-oriented programming
- Platform independence
- Strong security mechanisms
- Automatic memory management
- Exception handling
- Multithreading support
- Rich standard libraries

The Boarding Finder system uses Java to implement the application's business logic through Servlets, Data Access Objects (DAOs), model classes, and controller components.

2.5 Java Server Pages (JSP)

Java Server Pages (JSP) is a server-side technology used to create dynamic web pages. JSP combines HTML with Java code, allowing developers to generate dynamic content based on user requests and database information.

Compared with static HTML pages, JSP provides greater flexibility by enabling data retrieval from databases and displaying personalized information.

Some advantages of JSP include:

- Easy integration with Java Servlets
- Dynamic content generation
- Reusable components
- Faster development of web interfaces
- Support for Expression Language (EL) and JSTL

In the Boarding Finder project, JSP is used to develop the user interface, including:

- Home page
- Login page
- Registration page
- Search page
- Room details page
- Student dashboard
- Owner dashboard
- Add room page
- Edit room page

These pages communicate with Java Servlets to process user requests and display database information dynamically.

2.6 Java Servlets

Java Servlets are server-side Java programs that process client requests and generate responses. They act as controllers within the web application and manage communication between the user interface and the database.

When a user submits a form, the corresponding Servlet receives the request, validates the data, performs database operations, and returns the appropriate response.

The Boarding Finder system uses multiple Servlets to perform different operations, including:

- User Registration
- User Login
- Room Search
- Room Booking
- Booking Approval
- Booking Rejection
- Booking Cancellation
- Room Management
- Image Upload

This separation of responsibilities improves system maintainability and follows the Model-View-Controller (MVC) design pattern.

2.7 MySQL Database

MySQL is a relational database management system widely used in web application development. It stores data in tables and uses Structured Query Language (SQL) to retrieve and manipulate information.

The Boarding Finder system uses MySQL to store:

- User accounts
- Student information
- Owner information
- Boarding room details
- Booking records
- Room occupancy information

Using a relational database provides several advantages:

- Efficient data storage
- Data consistency
- Faster searching
- Reduced data redundancy
- Secure user authentication
- Reliable transaction management

MySQL integrates efficiently with Java through JDBC (Java Database Connectivity), enabling seamless communication between the application and the database.

2.8 Bootstrap Framework

Bootstrap is a popular front-end framework that simplifies the development of responsive and visually appealing websites. It provides pre-designed CSS classes and JavaScript components that help developers create modern user interfaces quickly.

Bootstrap includes:

- Responsive grid system
- Navigation bars
- Cards
- Buttons
- Forms
- Tables
- Alerts
- Modals

- Carousels

The Boarding Finder system uses Bootstrap to ensure that web pages display correctly on different screen sizes while maintaining a clean and professional appearance.

2.9 Apache Tomcat Server

Apache Tomcat is an open-source web server and servlet container developed by the Apache Software Foundation. It executes Java Servlets and JSP pages and provides the runtime environment for Java web applications.

Tomcat offers several advantages:

- Lightweight architecture
- Easy deployment
- High performance
- Strong security
- Wide community support

The Boarding Finder system was deployed and tested using Apache Tomcat Server, which handled all client requests and executed the application's server-side components.

2.10 Object-Oriented Programming (OOP)

Object-Oriented Programming is a software development methodology that organizes programs into reusable objects containing both data and methods.

The four fundamental principles of OOP are:

Encapsulation

Encapsulation protects data by restricting direct access to object attributes and allowing controlled access through methods.

Inheritance

Inheritance enables new classes to inherit properties and behaviors from existing classes, reducing code duplication.

Polymorphism

Polymorphism allows objects to perform different actions through the same interface, improving flexibility.

Abstraction

Abstraction hides unnecessary implementation details while exposing only essential functionalities to users.

The Boarding Finder system applies OOP principles through model classes such as **User**, **Student**, **Owner**, **Room**, and **Booking**, making the code modular, reusable, and easier to maintain.

2.11 Advantages of the Proposed System

Compared with traditional boarding management methods, the proposed Boarding Finder system offers several advantages:

- Online room searching from any location.
- Secure user registration and login.
- Advanced search filters for university, gender, location, and rental fee.
- Detailed room information with images and facilities.
- Online booking request submission.
- Separate dashboards for students and owners.
- Automatic room occupancy management.
- Prevention of duplicate bookings.

- Google Maps integration for location viewing.
- Modern responsive user interface with Dark Mode support.

These features improve efficiency, reduce manual work, and enhance the overall user experience.

2.12 Chapter Summary

This chapter reviewed the existing methods of boarding room management and discussed the technologies used to develop the Boarding Finder system. Traditional methods were found to be inefficient due to manual processes and limited accessibility. Modern web technologies such as Java, JSP, Servlets, MySQL, Bootstrap, and Apache Tomcat provide an effective foundation for developing a secure, reliable, and user-friendly boarding room management system.

The next chapter presents the system analysis and design, including the functional and non-functional requirements, system architecture, UML diagrams, database design, and overall system workflow.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

System analysis and design is an important phase in the Software Development Life Cycle (SDLC). It involves studying the requirements of the proposed system and designing a solution that satisfies user needs efficiently. A well-designed system improves maintainability, scalability, security, and overall performance.

The Boarding Finder system was designed to simplify the process of finding and managing boarding accommodations. The system provides two types of users: students and boarding owners. Students can search for boarding rooms, submit booking requests, and manage their bookings, while boarding owners can manage room information and booking requests through a dedicated dashboard.

This chapter presents the functional and non-functional requirements, feasibility study, system architecture, and system design models used in developing the Boarding Finder system.

3.2 Requirement Analysis

Requirement analysis is the process of identifying the functions and services that the system must provide to meet user expectations. Requirements are generally classified into two categories: functional requirements and non-functional requirements.

3.3 Functional Requirements

Functional requirements describe the specific services and operations that the system must perform.

The Boarding Finder system provides the following functional requirements.

User Registration

The system shall allow new users to register as either a student or a boarding owner. User information such as name, email address, password, telephone number, and user role shall be stored securely in the database.

User Login

The system shall authenticate registered users using their email address and password. After successful authentication, users shall be redirected to their respective dashboards according to their role.

Search Boarding Rooms

Students shall be able to search available boarding rooms using search filters such as:

- University
- Gender preference
- Maximum monthly rental fee

Only rooms that are currently available shall be displayed.

View Room Details

The system shall display detailed information for each boarding room including:

- Room images
- Monthly rental fee
- Boarding type
- Facilities
- Location

- Google Maps link
- Owner contact information
- Room availability

Booking Management

Students shall be able to submit booking requests for available rooms.

The system shall prevent duplicate booking requests for the same room.

Student Dashboard

Students shall be able to:

- View booking history
- View booking status
- Cancel bookings
- Track accepted and pending bookings

Owner Dashboard

Boarding owners shall be able to:

- Add new rooms
- Edit room information
- Delete rooms
- Upload room images
- View booking requests

- Accept bookings
- Reject bookings

Occupancy Management

The system shall automatically update room occupancy after an accepted booking.

When the room reaches its maximum capacity:

- Room status changes to **Booked**
- Remaining pending bookings are automatically rejected

When an accepted booking is cancelled:

- Occupancy decreases
- Room status changes back to **Available**

Theme Management

The system shall provide both Light Mode and Dark Mode interfaces.

3.4 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system.

Performance

The system should process user requests quickly with minimum response time.

Security

The system should authenticate users before allowing access to protected pages.

Passwords should be stored securely within the database.

Unauthorized users should not access restricted resources.

Reliability

The system should provide accurate room information and maintain booking consistency without data loss.

Availability

The application should be available whenever users require boarding information through an internet connection.

Usability

The user interface should be simple, attractive, and easy to understand for both students and boarding owners.

Scalability

The system should support future enhancements such as online payments, mobile applications, and notification services.

Maintainability

The software should follow Object-Oriented Programming principles to simplify future maintenance and upgrades.

3.5 Feasibility Study

A feasibility study evaluates whether the proposed system can be developed successfully.

3.5.1 Technical Feasibility

The Boarding Finder system was developed using widely available technologies including Java, JSP, Servlets, MySQL, Bootstrap, and Apache Tomcat.

These technologies are open-source, reliable, and supported by extensive documentation, making the project technically feasible.

3.5.2 Economic Feasibility

The development cost of the system is relatively low because all software tools used are free or open-source.

Examples include:

- NetBeans IDE
- Apache Tomcat
- MySQL
- XAMPP
- Bootstrap

Therefore, no significant software licensing costs are required.

3.5.3 Operational Feasibility

The system provides a user-friendly interface requiring minimal training.

Students and boarding owners can easily perform their tasks using standard web browsers.

3.5.4 Schedule Feasibility

The project was completed within the allocated academic project duration by following the Software Development Life Cycle.

3.6 Software Development Methodology

The Boarding Finder system was developed using the Waterfall Software Development Model.

The Waterfall model consists of several sequential phases.

Requirement Analysis

Project requirements were collected by studying the problems faced by university students and boarding owners.

System Design

Database design, UML diagrams, and software architecture were prepared before implementation.

Implementation

The application was developed using Java Server Pages, Java Servlets, MySQL, Bootstrap, HTML, CSS, and JavaScript.

Testing

Each module was tested individually before integrating the complete system.

Testing included:

- User registration
- Login
- Room searching
- Booking
- Booking approval
- Room management

Deployment

The application was deployed using Apache Tomcat Server.

Maintenance

Future improvements can be added without changing the entire system architecture.

3.7 System Architecture

The Boarding Finder system follows a three-tier architecture.

Presentation Layer

The presentation layer consists of JSP pages that interact directly with users.

Examples include:

- Home Page
- Login Page
- Registration Page

- Search Page
- Room Details
- Student Dashboard
- Owner Dashboard

Business Logic Layer

The business logic layer contains Java Servlets responsible for processing requests.

Examples include:

- Login Servlet
- Register Servlet
- Booking Servlet
- Search Servlet
- Room Servlet

This layer validates user input and performs business operations.

Data Layer

The data layer consists of MySQL database tables accessed through DAO classes using JDBC.

Examples include:

- UserDAO
- RoomDAO
- BookingDAO

This layer manages database communication.

3.8 Advantages of the Proposed Design

The proposed system architecture provides several advantages.

- Separation of presentation, business logic, and database layers.
- Easy maintenance.
- Improved security.
- Better scalability.
- Reduced code duplication.
- Faster development.
- Easy database management.
- Object-oriented implementation.

3.9 Chapter Summary

This chapter discussed the analysis and design of the Boarding Finder system. It presented the system requirements, feasibility analysis, software development methodology, and overall architecture. The design ensures that the system is modular, maintainable, secure, and capable of meeting the requirements of students and boarding owners. The following section of this chapter will present the UML diagrams and database design that describe the system structure and interactions in greater detail.

3.10 Use Case Diagram

A Use Case Diagram illustrates the interactions between the users (actors) and the system. It identifies the major functionalities available to each type of user.

The Boarding Finder system has two primary actors:

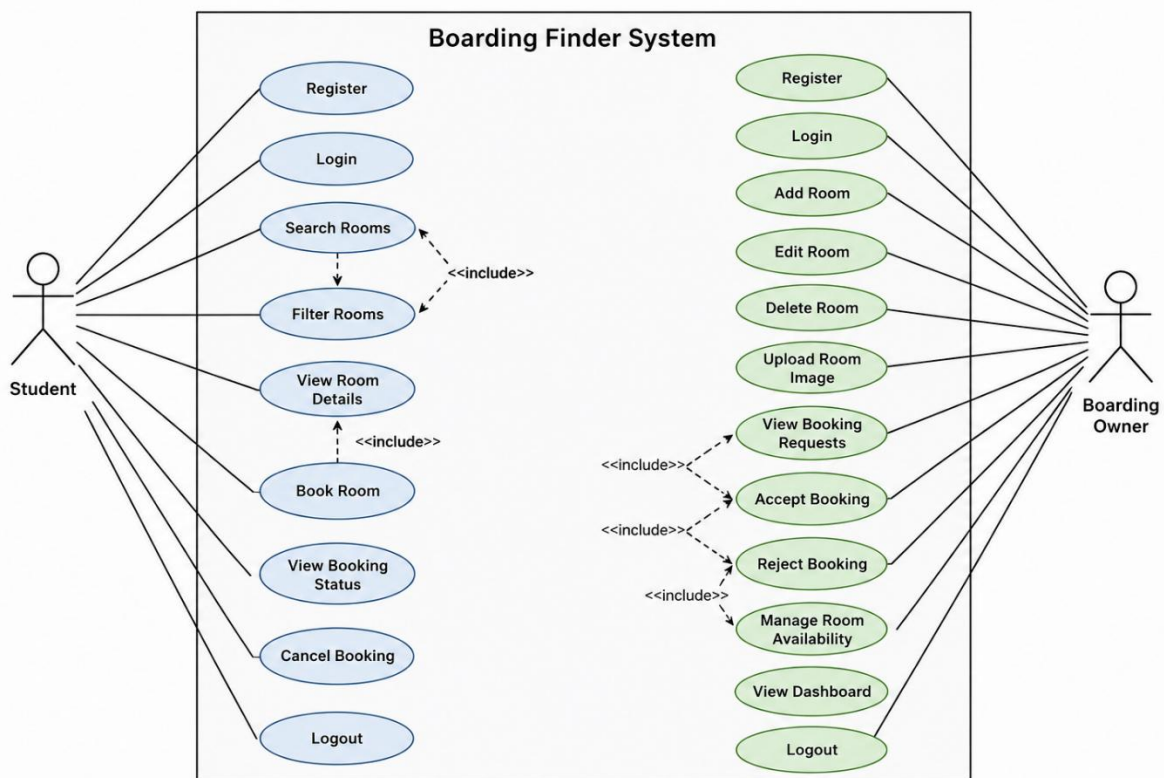
- Student

- Boarding Owner

The student can register, log in, search rooms, view room details, submit booking requests, monitor booking status, and cancel bookings.

The boarding owner can register, log in, manage room information, upload room images, review booking requests, accept or reject bookings, and monitor room occupancy.

Figure 3.1: Use Case Diagram of the Boarding Finder System

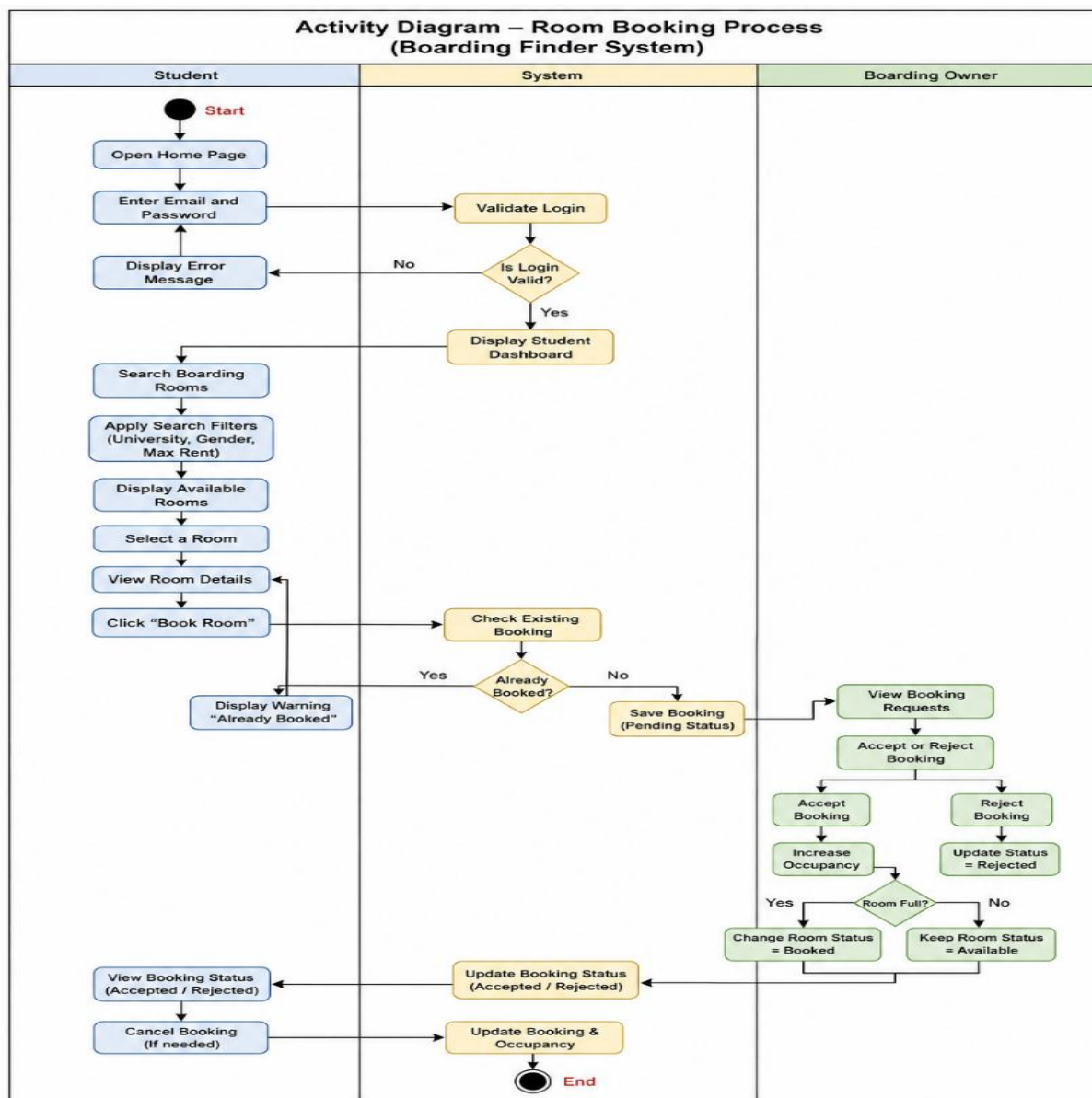


3.11 Use Case Descriptions

Use Case 1: User Registration

Use Case Name	User Registration
Primary Actor	Student / Boarding Owner
Description	Allows a new user to create an account.
Precondition	User is not already registered.

Main Flow	Enter details → Validate input → Save data → Registration successful.
Postcondition	User account is created successfully.



Use Case 2: User Login

Item	Description
Primary Actor	Student / Boarding Owner
Description	Authenticates users and redirects them to the appropriate dashboard.
Preconditions	User has a valid account.
Main Flow	Enter email and password → Validate credentials → Login successful.
Postcondition	User dashboard is displayed.

Use Case 3: Search Boarding Rooms

Item	Description
Primary Actor	Student
Description	Searches available boarding rooms using selected filters.
Preconditions	Student is logged in.
Main Flow	Select filters → Search → Display available rooms.
Postcondition	Matching rooms are displayed.

Use Case 4: Booking a Room

Item	Description
Primary Actor	Student
Description	Allows a student to submit a booking request.
Preconditions	Room must be available.
Main Flow	Open room details → Click Book → Submit booking request.
Postcondition	Booking request is stored with Pending status.

Use Case 5: Manage Rooms

Item	Description
Primary Actor	Boarding Owner
Description	Allows owners to add, edit, and delete boarding rooms.
Preconditions	Owner is logged in.
Main Flow	Add/Edit/Delete room → Save changes.
Postcondition	Room information is updated.

Use Case 6: Manage Bookings

Item	Description
Primary Actor	Boarding Owner
Description	Accept or reject booking requests.
Preconditions	Pending booking exists.
Main Flow	Review booking → Accept/Reject → Update booking status.
Postcondition	Booking status changes accordingly.

3.12 Activity Diagram

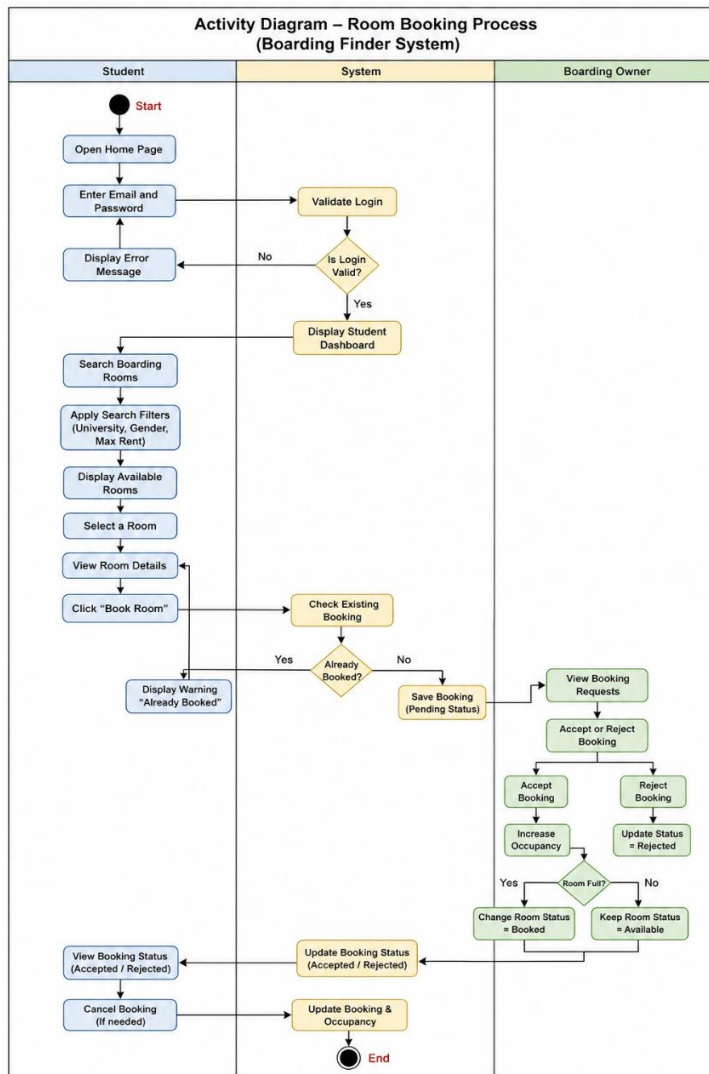
An Activity Diagram illustrates the workflow of the system from the beginning of a process to its completion.

For the Boarding Finder system, the main activities include:

- User Registration
- User Login
- Search Rooms
- View Room Details
- Submit Booking
- Owner Reviews Booking

- Booking Accepted or Rejected
- Student Cancels Booking (if necessary)

Figure 3.2: Activity Diagram of Room Booking Process



Activity Diagram Explanation

The activity begins when the student logs into the system. The student searches for boarding rooms using available filters. After selecting a room, the system displays detailed room information. If the room is suitable, the student submits a booking request. The owner reviews the request and either accepts or rejects it. If accepted, the occupancy count is updated automatically. If the room reaches its maximum capacity, its status changes to **Booked**.

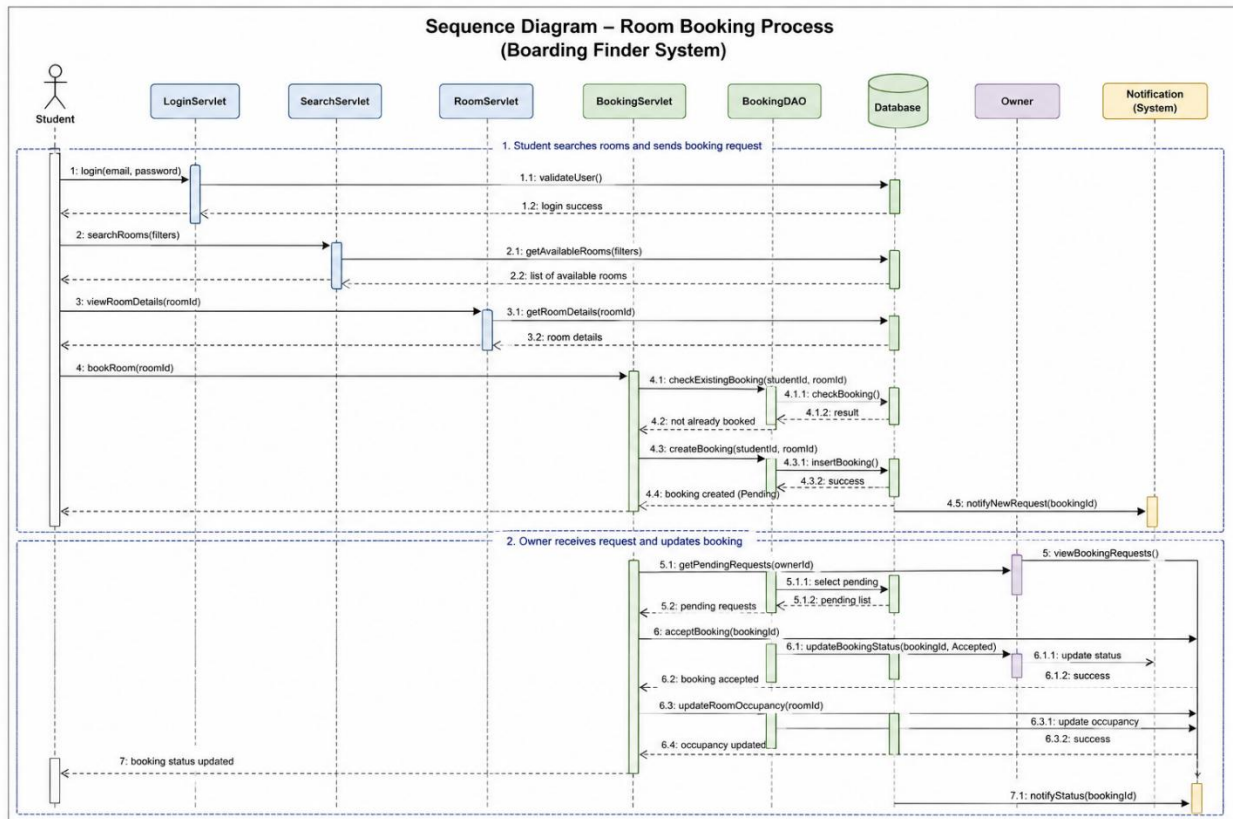
3.13 Sequence Diagram

A Sequence Diagram shows how different system components interact over time.

The sequence of operations for booking a room is as follows:

1. Student logs in.
2. Student searches available rooms.
3. Student opens room details.
4. Student submits booking request.
5. Booking Servlet validates request.
6. BookingDAO stores booking.
7. Database confirms booking.
8. Owner views request.
9. Owner accepts booking.
10. Occupancy count is updated.
11. Student views updated booking status.

Figure 3.3: Sequence Diagram for Room Booking



3.14 Class Diagram

The Boarding Finder system follows Object-Oriented Programming principles. The Class Diagram represents the structure of the classes and their relationships.

The main classes include:

- User
- Student
- Owner
- Room
- Booking

- UserDao
- RoomDAO
- BookingDAO
- DBConnection

Each class contains its own attributes and methods.

Examples:

User

Attributes:

- userId
- fullName
- email
- password
- role

Methods:

- login()
- register()

Room

Attributes:

- roomId
- title
- university

- location
- rent
- capacity
- occupied
- status

Methods:

- addRoom()
- updateRoom()
- deleteRoom()

Booking

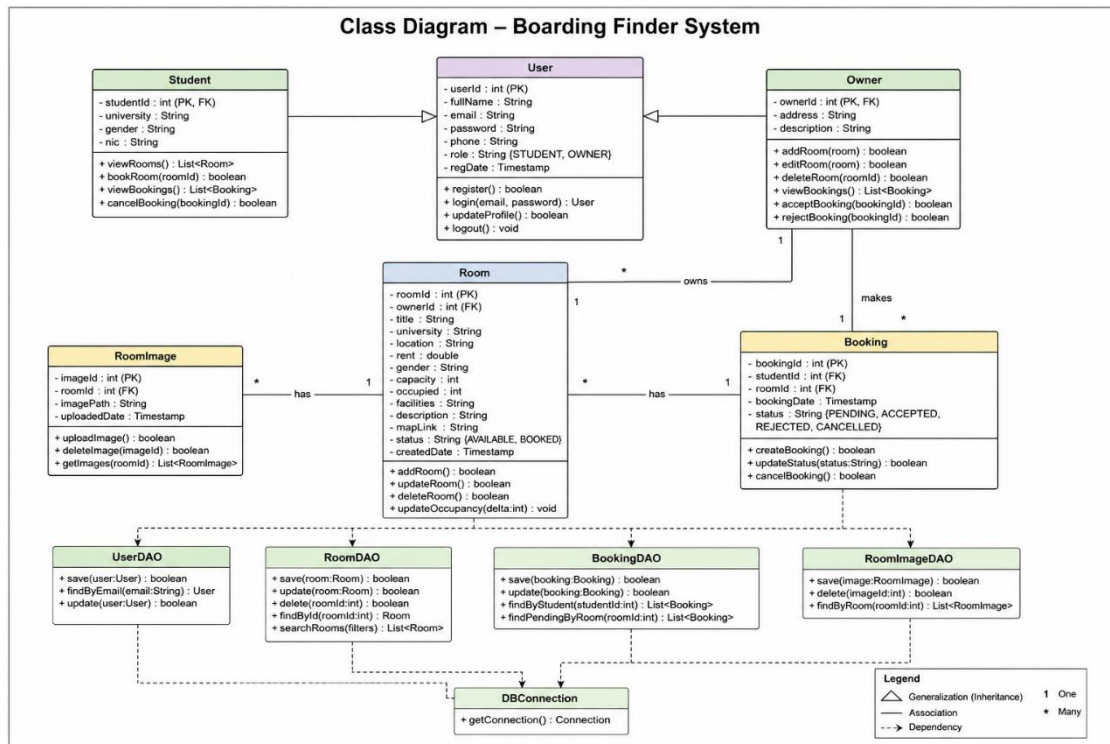
Attributes:

- bookingId
- bookingDate
- bookingStatus

Methods:

- createBooking()
- cancelBooking()
- updateStatus()

Figure 3.4: Class Diagram of the Boarding Finder System



3.15 Database Design

The Boarding Finder system uses a relational database implemented in MySQL.

The database stores:

- User information
- Room information
- Booking details

The relationships between these entities ensure data consistency and minimize redundancy.

3.16 Entity Relationship (ER) Diagram

The ER Diagram illustrates the relationships among the database tables.

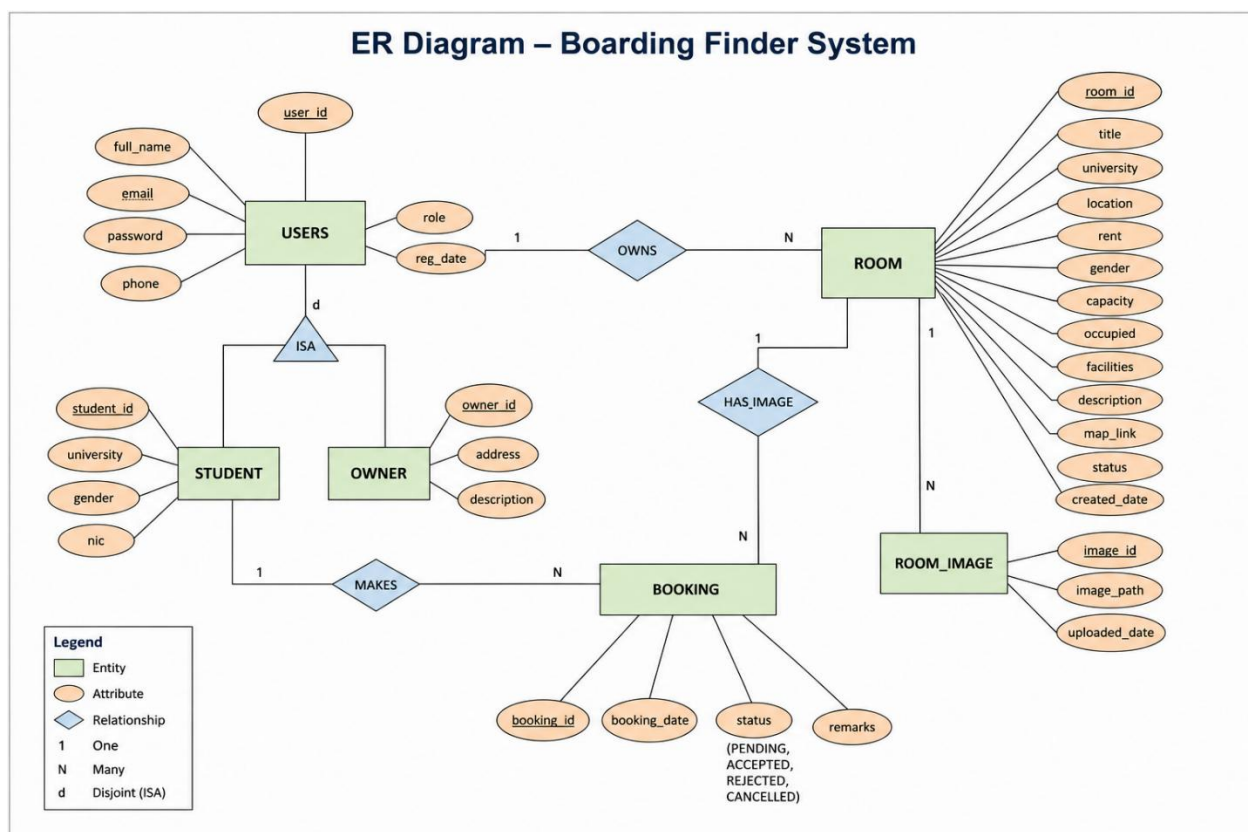
The major entities are:

- User
- Room
- Booking

Relationships:

- One owner can manage many rooms.
- One student can create many bookings.
- One room can receive many booking requests.
- One booking belongs to one student and one room.

Figure 3.5: Entity Relationship Diagram



3.17 Data Dictionary

Table: Users

Field Name	Data Type	Description
user_id	INT	Primary Key
full_name	VARCHAR(100)	User's full name
email	VARCHAR(100)	Email address
password	VARCHAR(255)	Password
phone	VARCHAR(20)	Contact number
role	VARCHAR(20)	Student or Owner

Table: Rooms

Field Name	Data Type	Description
room_id	INT	Primary Key
owner_id	INT	Foreign Key
title	VARCHAR(100)	Room title
university	VARCHAR(100)	Nearby university
location	VARCHAR(100)	Boarding location
rent	DECIMAL	Monthly rental fee
capacity	INT	Maximum capacity

Table: Bookings

Field Name	Data Type	Description
booking_id	INT	Primary Key
student_id	INT	Foreign Key
room_id	INT	Foreign Key
booking_date	DATE	Booking date
status	VARCHAR(30)	Pending, Accepted, Rejected, Cancelled

3.18 Chapter Summary

This chapter presented the analysis and design of the Boarding Finder system. It discussed the system requirements, feasibility study, software development methodology, and software architecture. Furthermore, it described the UML diagrams and database design used to model the system before implementation. These design artifacts provided a clear blueprint for developing a reliable, maintainable, and user-friendly web application.

3.9 Chapter Summary

This chapter discussed the analysis and design of the Boarding Finder system. It presented the system requirements, feasibility analysis, software development methodology, and overall architecture. The design ensures that the system is modular, maintainable, secure, and capable of meeting the requirements of students and boarding owners. The following section of this chapter will present the UML diagrams and database design that describe the system structure and interactions in greater detail.

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 Introduction

This chapter describes the implementation of the Boarding Finder System. The system was developed as a web-based application using Java, Java Server Pages (JSP), Servlets, MySQL, HTML, CSS, JavaScript, and Bootstrap. The implementation process involved developing individual modules, integrating them into a complete system, and testing each functionality to ensure correct operation.

The Boarding Finder System consists of two main user roles: students and boarding owners. Students can search for boarding rooms, view room details, submit booking requests, and manage their bookings. Boarding owners can manage boarding rooms, upload room images, and handle booking requests through a dedicated dashboard. The system also includes automatic room occupancy management, duplicate booking prevention, Google Maps integration, and dark/light mode support.

4.2 Development Environment

The Boarding Finder System was developed using the following software tools and technologies.

Table 4.1: Development Environment

4.3 System Modules

The Boarding Finder System was divided into several modules to improve maintainability and simplify development.

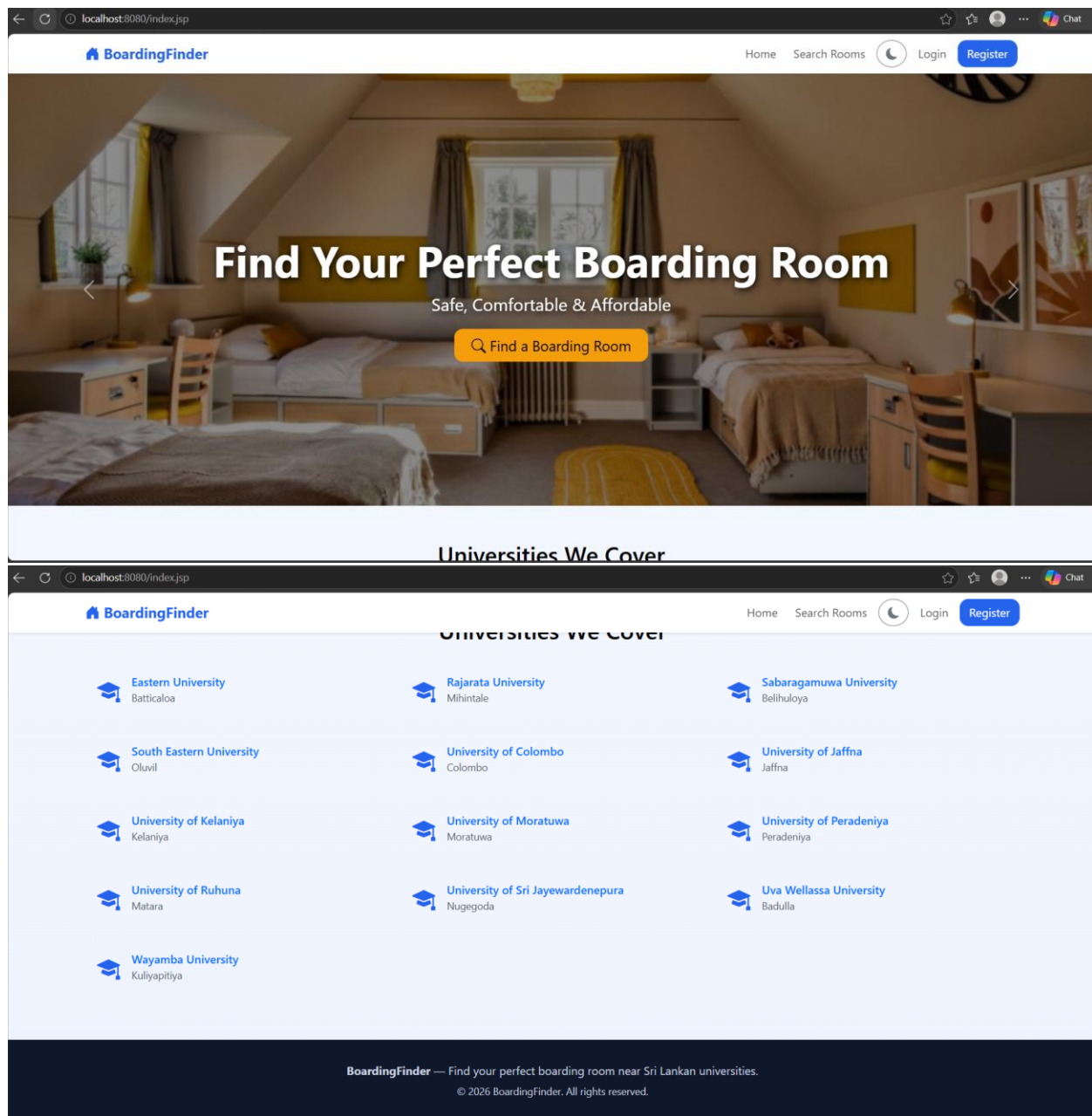
4.3.1 Home Page Module

The Home Page serves as the main entry point to the application. It introduces the Boarding Finder System and provides navigation links to the major functionalities of the system.

The homepage contains:

- Navigation bar
- Bootstrap image carousel
- University section
- Search button
- Login button
- Registration button
- Footer

Figure 4.1: Home Page



4.3.2 User Registration Module

The registration module allows new users to create accounts as either students or boarding owners.

The registration form collects:

- Full name

- Email address
- Password
- Telephone number
- User role

Before saving the data, the system validates user input to ensure that required fields are completed and duplicate email addresses are not registered.

After successful registration, user information is stored securely in the MySQL database.

Figure 4.2: User Registration Page

4.3.3 User Login Module

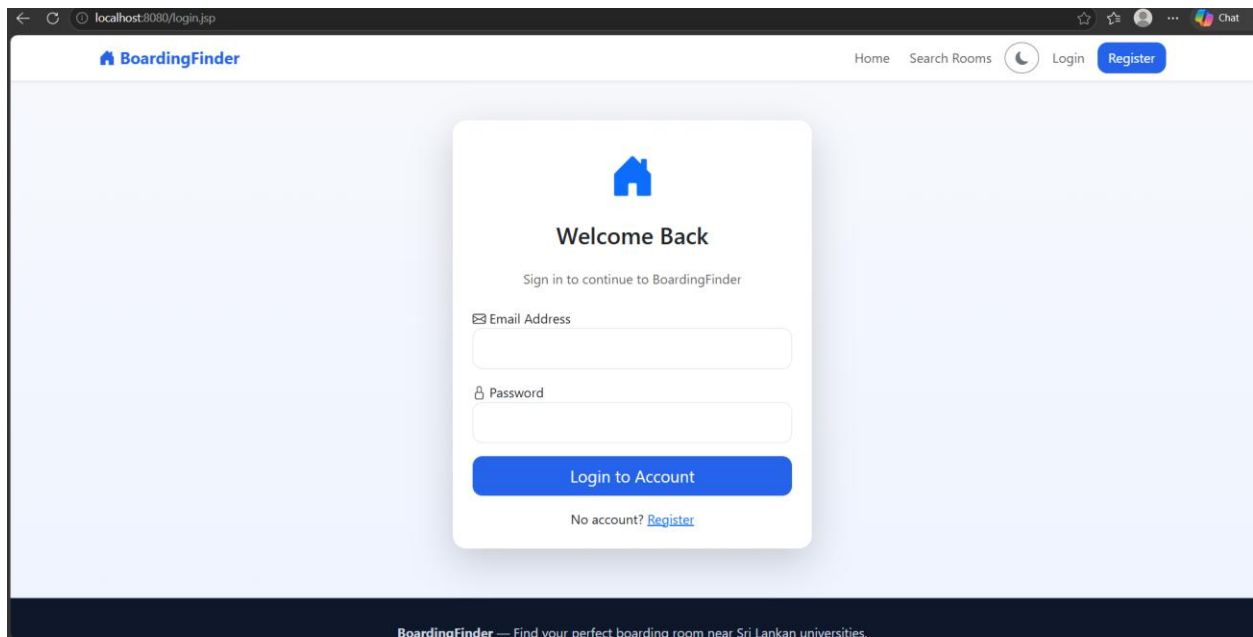
The login module authenticates registered users using their email address and password.

Depending on the user role, the system redirects users to the appropriate dashboard:

- Student Dashboard
- Boarding Owner Dashboard

If incorrect login credentials are entered, the system displays an appropriate error message.

Figure 4.3: Login Page



4.3.4 Search Boarding Rooms Module

Students can search for available boarding rooms using multiple search filters.

Available search criteria include:

- University
- Gender preference
- Maximum monthly rent

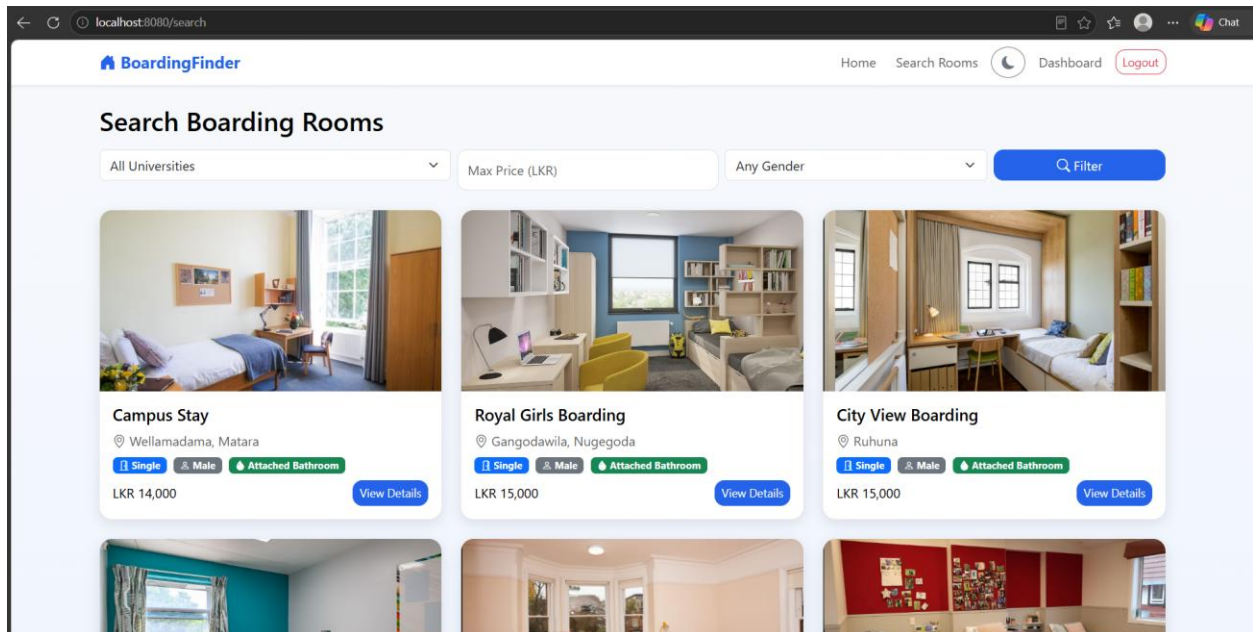
Only rooms with available capacity are displayed.

The search results include:

- Room image
- Monthly rent
- Boarding location
- Availability status

- View Details button

Figure 4.4: Search Results Page



4.3.5 Room Details Module

The Room Details page displays complete information about the selected boarding room.

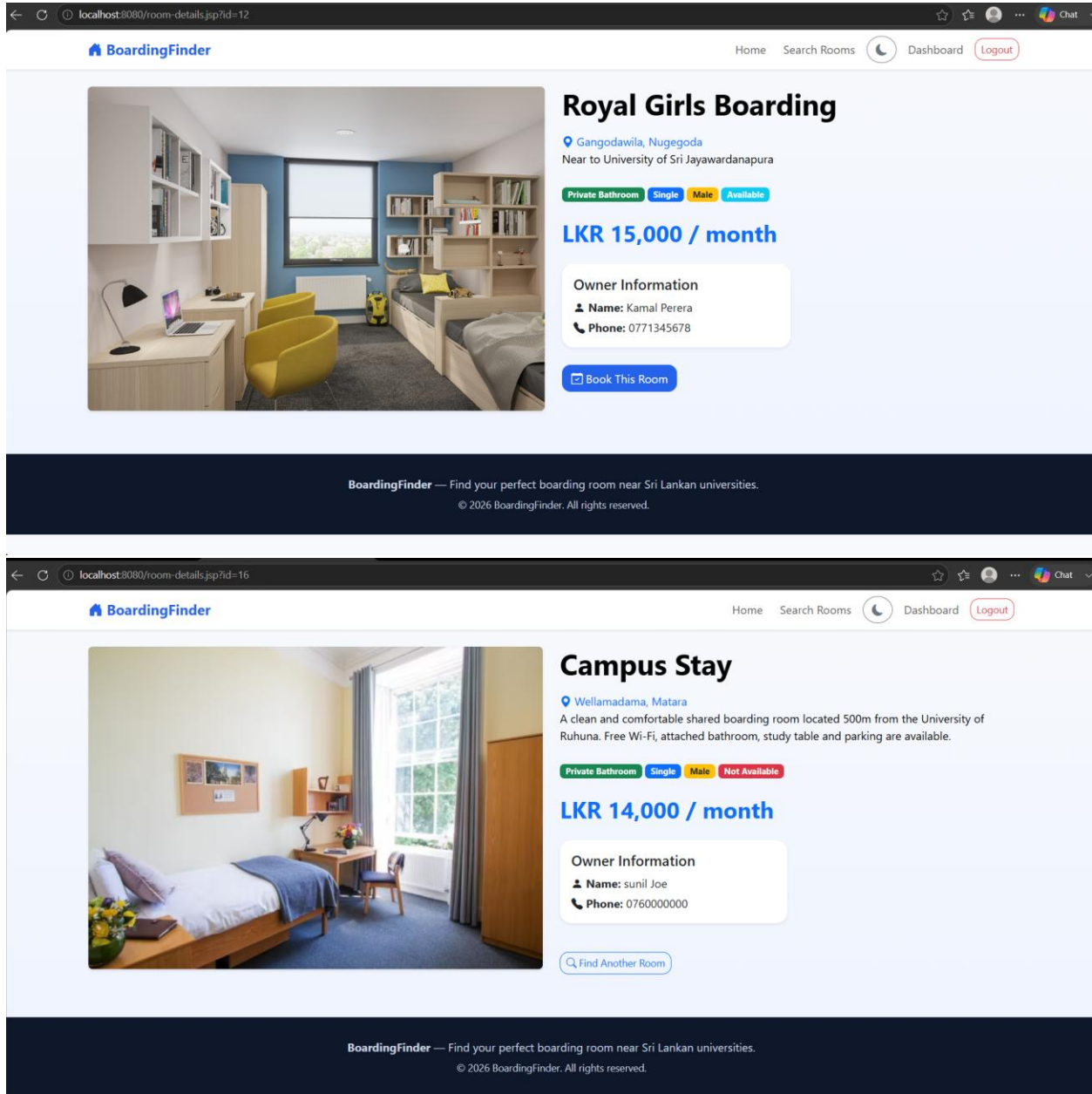
Information includes:

- Room photographs
- Monthly rental fee
- Boarding facilities
- Boarding location
- Google Maps link
- Owner contact information
- Room availability

- Book Room button

This module enables students to make informed booking decisions.

Figure 4.5: Room Details Page



4.3.6 Booking Module

The Booking Module enables students to submit booking requests for available boarding rooms.

The booking process includes several validation steps.

The system:

- Prevents duplicate booking requests.
- Stores new bookings with a **Pending** status.
- Automatically updates booking status after the owner's decision.

After submission, students can monitor booking progress through their dashboard.

Figure 4.6: Booking Page

The image shows two screenshots of the BoardingFinder web application. The top screenshot displays a room listing for 'Royal Girls Boarding' located in Gangodawila, Nugegoda, near the University of Sri Jayawardanapura. The room features a private bathroom, is single occupancy, and is available for males. The monthly fee is LKR 15,000. The owner's name is Kamal Perera, and the contact number is 0771345678. A 'Book This Room' button is visible. The bottom screenshot shows a student dashboard for 'Ilma!' with a green notification bar stating 'Booking request sent successfully!'. Below the notification, there is a 'Welcome, Ilma!' message and a 'Find More Rooms' button. A table lists the booking details:

#	Room	Booked On	Status	Action
1	Royal Girls Boarding	2026-07-25 08:44:22.0	Pending	Cancel

4.3.7 Student Dashboard

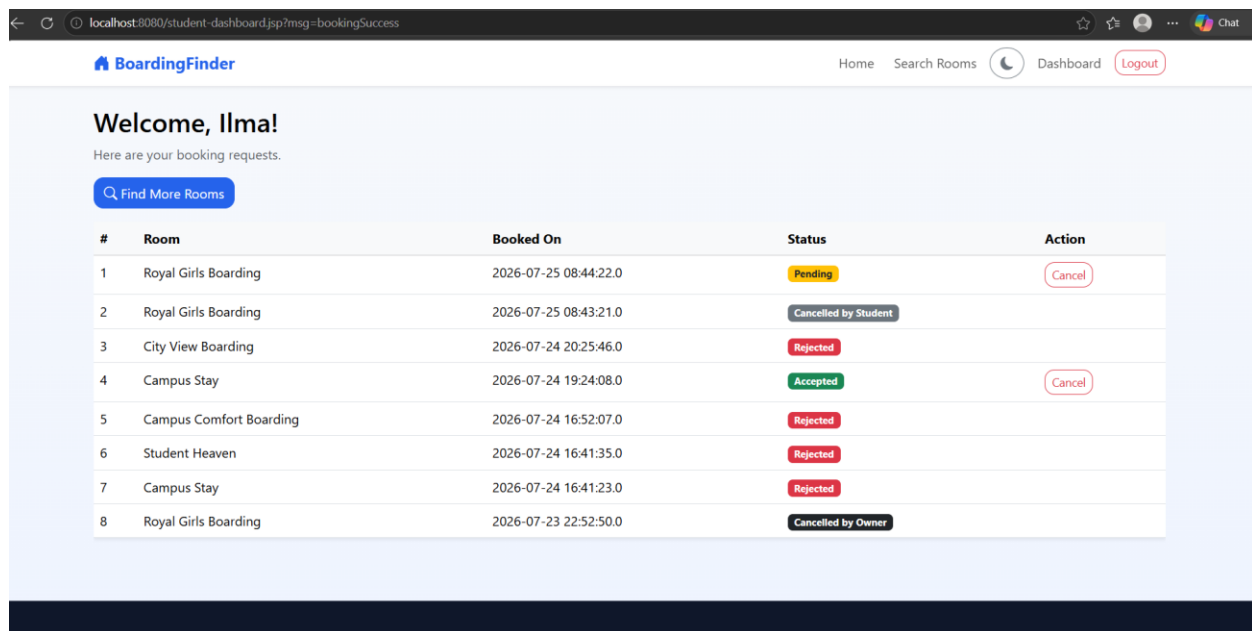
The Student Dashboard provides a personalized interface for students to manage their bookings.

Students can:

- View current bookings
- Check booking status
- Cancel accepted bookings
- Review rejected bookings
- View room information

The dashboard provides a centralized view of all student activities.

Figure 4.7: Student Dashboard



The screenshot shows the Student Dashboard in the BoardingFinder application. The dashboard includes a navigation bar with links to Home, Search Rooms, Dashboard, and Logout. A welcome message 'Welcome, Ilma!' is displayed, followed by a button to 'Find More Rooms'. Below this is a table of booking requests.

#	Room	Booked On	Status	Action
1	Royal Girls Boarding	2026-07-25 08:44:22.0	Pending	Cancel
2	Royal Girls Boarding	2026-07-25 08:43:21.0	Cancelled by Student	
3	City View Boarding	2026-07-24 20:25:46.0	Rejected	
4	Campus Stay	2026-07-24 19:24:08.0	Accepted	Cancel
5	Campus Comfort Boarding	2026-07-24 16:52:07.0	Rejected	
6	Student Heaven	2026-07-24 16:41:35.0	Rejected	
7	Campus Stay	2026-07-24 16:41:23.0	Rejected	
8	Royal Girls Boarding	2026-07-23 22:52:50.0	Cancelled by Owner	

4.3.8 Boarding Owner Dashboard

The Boarding Owner Dashboard allows owners to manage boarding room information and booking requests.

Owners can:

- Add new rooms
- Edit room details
- Delete rooms
- Upload room images
- View booking requests
- Accept bookings
- Reject bookings

This dashboard provides efficient room management through a simple interface.

Figure 4.8: Owner Dashboard

The screenshot displays the BoardingFinder Owner Dashboard. The top navigation bar includes links for Home, Search Rooms, Dashboard, and Logout. The main content area is divided into two sections: 'Your Rooms' and 'Booking Requests'.

Your Rooms Section: This section features three room listings, each with a photo, name, price, status, and edit/delete buttons. A '+ Add Room' button is located in the top right corner.

Room Name	Price (LKR)	Status
Campus Comfort Boarding	20,000	Available
Student Heaven	18,000	Booked
Campus Stay	14,000	Booked

Booking Requests Section: This section contains a table listing individual booking requests with columns for ID, Student, Room, Date, Status, and Actions.

#	Student	Room	Date	Status	Actions
1	Hansi Kaveena	Campus Comfort Boarding	2026-07-24 21:12:35.0	Cancelled by Owner	
2	Ilma Yasmin	Campus Stay	2026-07-24 19:24:08.0	Accepted	Cancel Booking
3	Hansi Kaveena	Campus Stay	2026-07-24 19:23:38.0	Cancelled by Student	
4	Hansi Kaveena	Campus Comfort Boarding	2026-07-24 17:29:10.0	Cancelled by Student	
5	Hansi Kaveena	Campus Stay	2026-07-24 17:03:54.0	Cancelled by Student	
6	Ilma Yasmin	Campus Comfort Boarding	2026-07-24 16:52:07.0	Rejected	
7	Ilma Yasmin	Student Heaven	2026-07-24 16:41:35.0	Rejected	
8	Ilma Yasmin	Campus Stay	2026-07-24 16:41:23.0	Rejected	
9	Hansi Kaveena	Student Heaven	2026-07-24 16:30:07.0	Accepted	Cancel Booking

The footer of the dashboard reads: 'BoardingFinder — Find your perfect boarding room near Sri Lankan universities.'

4.3.9 Room Management Module

Boarding owners can manage room information through CRUD operations.

The system supports:

- Add Room
- Edit Room
- Delete Room

Each room record contains:

- Room title
- University
- Monthly rent
- Capacity
- Occupancy
- Description
- Facilities
- Location
- Google Maps link

The updated information becomes immediately available to students.

Figure 4.9: Room Management

The screenshot displays the 'Edit Room' form in the BoardingFinder application. The form is titled 'Edit Room' and contains the following fields:

- Title:** Campus Comfort Boarding
- Description:** Affordable shared boarding with free Wi-Fi, laundry facilities and secure parking. Suitable for university students.
- Location:** Peradeniya, Kandy
- Price (LKR):** 20000.0
- Room Type:** Single
- Gender:** Male
- Current Image:** A thumbnail image of a room with two beds. Below the thumbnail is a file upload area with a 'Choose File' button and 'No file chosen' text.

A green 'Update Room' button is located at the bottom of the form.

4.3.10 Image Upload Module

Owners can upload photographs of boarding rooms.

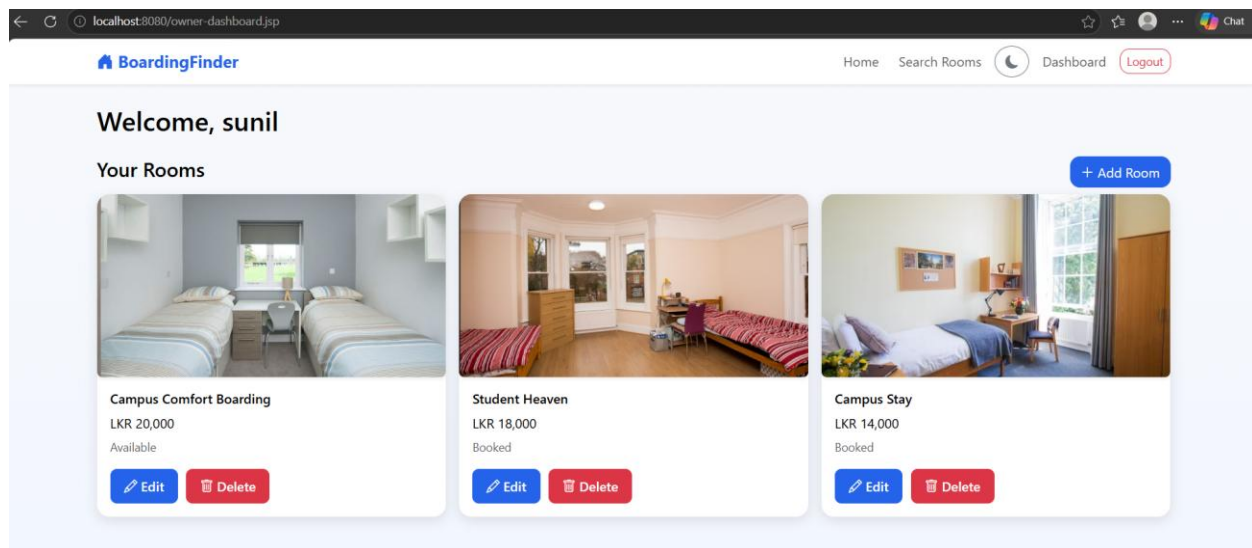
Uploaded images are stored on the server, while image paths are saved in the database.

The images are displayed on:

- Search Results
- Room Details

This feature helps students evaluate boarding rooms visually.

Figure 4.10: Image Upload



4.3.11 Booking Approval Module

After students submit booking requests, owners review each request.

Owners may:

- Accept booking
- Reject booking

When a booking is accepted:

- Occupancy increases automatically.
- Room availability is updated.

When a booking is rejected:

- Booking status changes to **Rejected**.

Students can immediately view the updated status.

Figure 4.11: Booking Approval

Booking Requests					
#	Student	Room	Date	Status	Actions
1	Ilma Yasmin	Campus Stay	2026-07-25 08:53:16.0	Pending	Accept Reject
2	Hadi Kusuma	Campus Comfort Reading	2026-07-24 21:13:25.0	Cancelled by Owner	
Booking Requests					
#	Student	Room	Date	Status	Actions
1	Ilma Yasmin	Campus Stay	2026-07-25 08:53:16.0	Accepted	Cancel Booking

4.3.12 Capacity Management Module

The Boarding Finder System automatically manages room occupancy.

Each room maintains:

- Maximum capacity
- Current occupancy

When the room reaches its maximum capacity:

- Room status changes from **Available** to **Booked**.
- Remaining pending booking requests are automatically rejected.

If a student cancels an accepted booking:

- Occupancy decreases.
- Room status changes back to **Available**.

This automatic process eliminates manual updates by boarding owners.

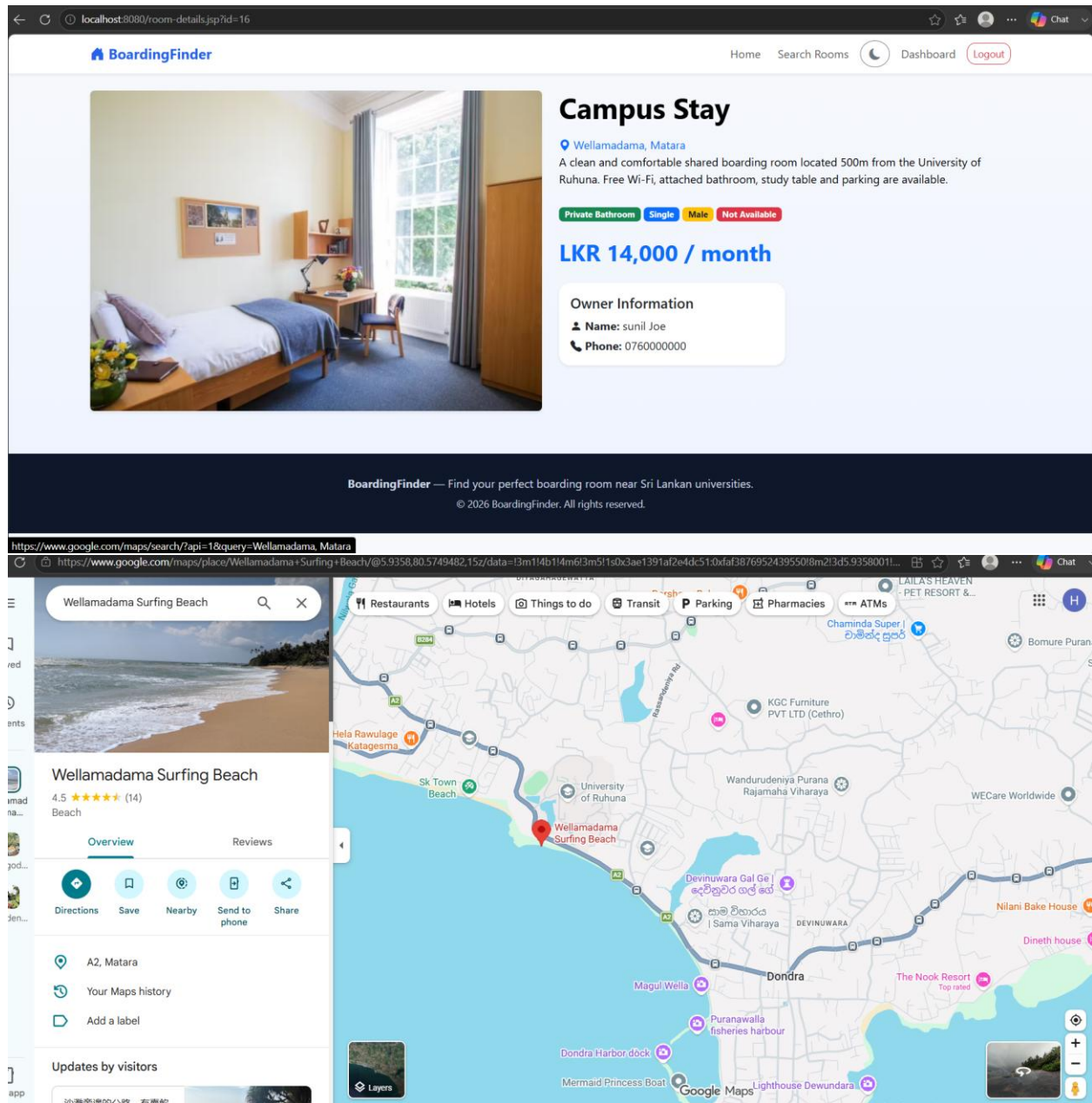
4.3.13 Google Maps Integration

The Boarding Finder System integrates Google Maps to improve location visibility.

Each room contains a Google Maps link that enables students to view the exact boarding location before making a booking.

This feature assists students in selecting accommodation close to their university.

Figure 4.12: Google Maps Integration



4.3.14 Dark Mode and Light Mode

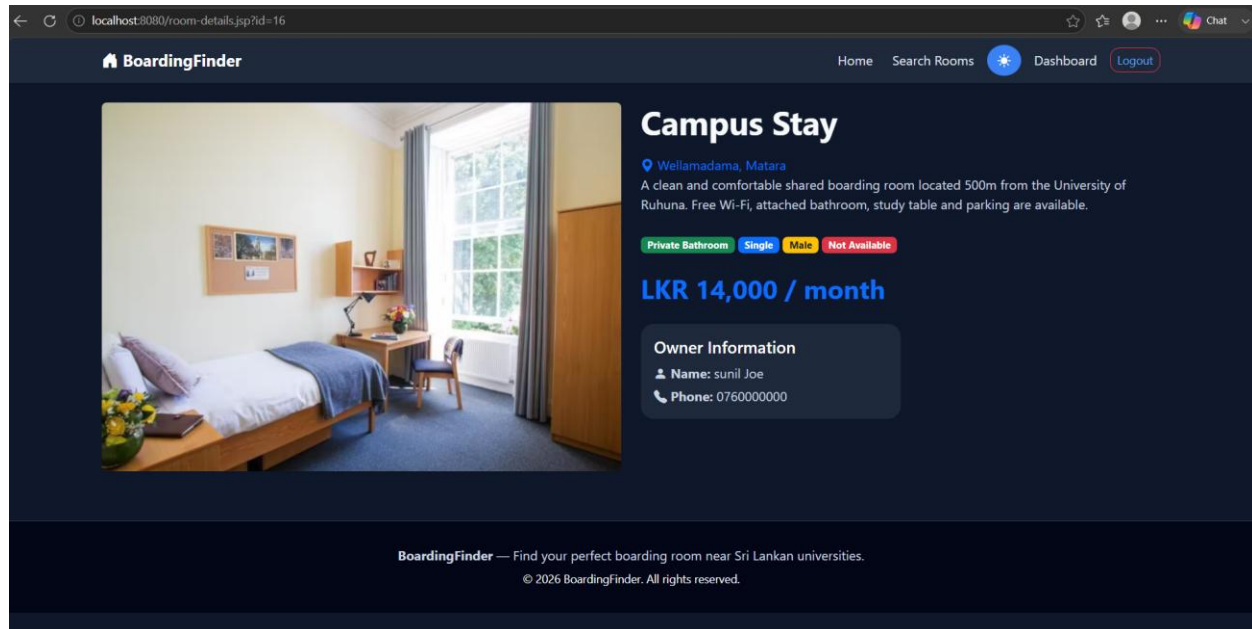
The system provides both Light Mode and Dark Mode interfaces.

Users can switch between themes according to their preferences.

Dark Mode reduces eye strain during nighttime usage, while Light Mode provides a traditional interface suitable for daytime use.

The selected theme improves user experience and accessibility.

Figure 4.13: Dark Mode Interface



4.4 Database Implementation

The Boarding Finder System uses a MySQL relational database to store all application data.

The database consists of several interconnected tables, including:

- Users
- Rooms
- Bookings
- Room Images

Primary keys and foreign keys are used to maintain referential integrity and ensure consistency among the tables.

Figure 4.14: Database Tables

Table	Action	Rows	Type	Collation	Size	Overhead
☐ bookings	Browse Structure Search Insert Empty Drop	23	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
☐ rooms	Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_unicode_ci	64.0 KiB	-
☐ universities	Browse Structure Search Insert Empty Drop	13	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
☐ users	Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
4 tables	Sum	52	InnoDB	utf8mb4_unicode_ci	160.0 KiB	0 B

		id	first_name	last_name	email	phone	password	role
☐	Edit Copy Delete	1	Kasun	Perera	owner@test.com	0771111111	1234	Owner
☐	Edit Copy Delete	2	Nimal	Silva	student@test.com	0772222222	1234	Student
☐	Edit Copy Delete	3	Hansi	Kaveena	hansiholuwigodage2003@gmail.com	0771234567	12345	Student
☐	Edit Copy Delete	4	Kamal	Perera	kamal@gmail.com	0771345678	kamal	Owner
☐	Edit Copy Delete	5	sunil	Joe	sunil@gmail.com	0760000000	sunil	Owner
☐	Edit Copy Delete	6	ilma	Yasmin	ilma@gmail.com	0765678763	ilma123	Student
☐	Edit Copy Delete	7	Kasuni	Perera	kasuni@gmail.com	0778964535	12345	Owner
☐	Edit Copy Delete	8	Ravichandran	Praveena	praveena@gmail.com	0786543234	12345	Student
☐	Edit Copy Delete	9	Hansi	Kaveena	hansi@gmail.com	0715364987	12345	Student
☐	Edit Copy Delete	10	saman	Perera	saman@gmail.com	0771234567	saman	Student

☐ Check all With selected: Edit Copy Delete Export

		university_id	university_name	city
☐	Edit Copy Delete	1	University of Colombo	Colombo
☐	Edit Copy Delete	2	University of Peradeniya	Peradeniya
☐	Edit Copy Delete	3	University of Sri Jayewardenepura	Nugegoda
☐	Edit Copy Delete	4	University of Kelaniya	Kelaniya
☐	Edit Copy Delete	5	University of Moratuwa	Moratuwa
☐	Edit Copy Delete	6	University of Ruhuna	Matara
☐	Edit Copy Delete	7	University of Jaffna	Jaffna
☐	Edit Copy Delete	8	Eastern University	Batticaloa
☐	Edit Copy Delete	9	Rajarata University	Mihintale
☐	Edit Copy Delete	10	Sabaragamuwa University	Belihuloya
☐	Edit Copy Delete	11	Wayamba University	Kuliypitiya
☐	Edit Copy Delete	12	South Eastern University	Oluvil
☐	Edit Copy Delete	13	Uva Wellassa University	Badulla

room_id	owner_id	university_id	title	description	location	room_type	gender	price	bathroom	image	status	capacity	occupied
12	4	3	Royal Girls Boarding	Near to University of Sri Jayawardanapura	Gangodawila, Nugegoda	Single	Male	15000.00	1	1784802873057_Screenshot 2026-07-21 224942.png	Available	1	0
14	4	6	City View Boarding	Near to University of Ruhuna	Ruhuna	Single	Male	15000.00	1	1784888898426_Screenshot 2026-07-23 163427.png	Booked	1	1
16	5	6	Campus Stay	A clean and comfortable shared boarding room located...	Wellamadama, Matara	Single	Male	14000.00	1	1784812063201_Screenshot 2026-07-23 183017.png	Booked	1	1
17	5	4	Student Heaven	Fully furnished single rooms with attached bathroom...	Dalugama, Kelaniya	Shared	Any	18000.00	1	1784812209368_Screenshot 2026-07-23 182936.png	Booked	1	1
18	5	2	Campus Comfort Boarding	Affordable shared boarding with free Wi-Fi, laundr...	Peradeniya, Kandy	Single	Male	20000.00	0	1784949654188_Screenshot 2026-07-24 221145.png	Available	1	0
19	7	5	Lake View Boarding	Clean and modern boarding house with attached bath...	Katubedda, Moratuwa	Single	Female	17000.00	0	1784813129434_Screenshot 2026-07-23 184944.png	Booked	1	1

←T→		booking_id	student_id	room_id	booking_date	booking_status
☐	Edit	Copy	Delete	7	3	12 2026-07-23 12:55:12 Rejected
☐	Edit	Copy	Delete	9	8	12 2026-07-23 14:43:36 Cancelled by Owner
☐	Edit	Copy	Delete	12	6	12 2026-07-23 17:22:50 Cancelled by Owner
☐	Edit	Copy	Delete	16	3	19 2026-07-24 05:52:31 Accepted
☐	Edit	Copy	Delete	17	3	14 2026-07-24 06:19:45 Cancelled by Owner
☐	Edit	Copy	Delete	18	3	12 2026-07-24 06:48:49 Rejected
☐	Edit	Copy	Delete	22	9	17 2026-07-24 11:00:07 Accepted
☐	Edit	Copy	Delete	23	9	12 2026-07-24 11:00:17 Cancelled by Owner
☐	Edit	Copy	Delete	24	6	16 2026-07-24 11:11:23 Rejected
☐	Edit	Copy	Delete	25	6	17 2026-07-24 11:11:35 Rejected
☐	Edit	Copy	Delete	27	6	18 2026-07-24 11:22:07 Rejected
☐	Edit	Copy	Delete	28	3	16 2026-07-24 11:33:54 Cancelled by Student
☐	Edit	Copy	Delete	29	3	18 2026-07-24 11:59:10 Cancelled by Student
☐	Edit	Copy	Delete	30	3	16 2026-07-24 13:53:38 Cancelled by Student
☐	Edit	Copy	Delete	31	6	16 2026-07-24 13:54:08 Cancelled by Owner
☐	Edit	Copy	Delete	32	6	14 2026-07-24 14:55:46 Rejected
☐	Edit	Copy	Delete	33	8	14 2026-07-24 14:56:15 Accepted
☐	Edit	Copy	Delete	34	3	18 2026-07-24 15:42:35 Cancelled by Owner
☐	Edit	Copy	Delete	35	3	12 2026-07-24 17:50:45 Cancelled by Student
☐	Edit	Copy	Delete	36	10	12 2026-07-24 18:08:22 Cancelled by Student
☐	Edit	Copy	Delete	37	6	12 2026-07-25 03:13:21 Cancelled by Student
☐	Edit	Copy	Delete	38	6	12 2026-07-25 03:14:22 Pending
Console	Edit	Copy	Delete	39	6	16 2026-07-25 03:23:16 Accepted

4.5 System Testing

The completed system was tested to verify that each module performed according to the specified requirements.

Test Case	Expected Result	Actual Result	Status
User Registration	Account created	Successful	Pass
User Login	Dashboard displayed	Successful	Pass
Search Rooms	Matching rooms displayed	Successful	Pass
View Room Details	Correct details displayed	Successful	Pass
Submit Booking	Booking saved	Successful	Pass
Duplicate Booking	Warning displayed	Successful	Pass
Accept Booking	Status updated	Successful	Pass
Reject Booking	Status updated	Successful	Pass
Cancel Booking	Booking cancelled	Successful	Pass
Add Room	Room added	Successful	Pass
Edit Room	Room updated	Successful	Pass
Delete Room	Room removed	Successful	Pass
Upload Image	Image uploaded	Successful	Pass
Google Maps	Location displayed	Successful	Pass
Dark Mode	Theme switched	Successful	Pass

Table 4.2: System Testing Results

4.6 Chapter Summary

This chapter presented the implementation of the Boarding Finder System. The system was successfully developed using Java, JSP, Servlets, MySQL, Bootstrap, HTML, CSS, and JavaScript. Each module was implemented according to the system requirements, including user registration, login, room searching, booking management, room management, image uploading, Google Maps integration, automatic capacity management, and theme switching. System testing confirmed that all implemented functionalities operated correctly and met the project objectives.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the results obtained from implementing the Boarding Finder System and discusses how the developed application satisfies the project objectives. The completed system was tested under different scenarios to evaluate its functionality, usability, and reliability. The discussion focuses on the implemented modules, system performance, and the benefits provided to students and boarding owners.

5.2 System Results

The Boarding Finder System was successfully implemented as a web-based application using Java, JSP, Servlets, MySQL, HTML, CSS, JavaScript, and Bootstrap. The system provides separate interfaces for students and boarding owners, allowing them to perform their respective tasks efficiently.

All major functions operated correctly during testing. Students were able to register, log in, search for boarding rooms, submit booking requests, monitor booking status, and cancel bookings. Boarding owners successfully managed room information, uploaded room images, and accepted or rejected booking requests through the owner dashboard.

The implemented system fulfilled all the functional requirements identified during the analysis phase.

5.3 Discussion of Implemented Modules

5.3.1 User Registration and Login

The registration and login modules worked successfully for both students and boarding owners. New users were able to create accounts by providing the required information, and registered users could access the system using their email addresses and passwords. Based on the selected role, users were redirected to the appropriate dashboard.

Figure 5.1: User Registration and Login

localhost:8080/register.jsp

BoardingFinder

HomeSearch RoomsLoginRegister

Create Your Account

Join BoardingFinder and discover boarding rooms near universities across Sri Lanka.

First Name

Last Name

Email Address

Phone Number

07XXXXXXXX

Password

Register As

Email Address

Phone Number

07XXXXXXXX

Password

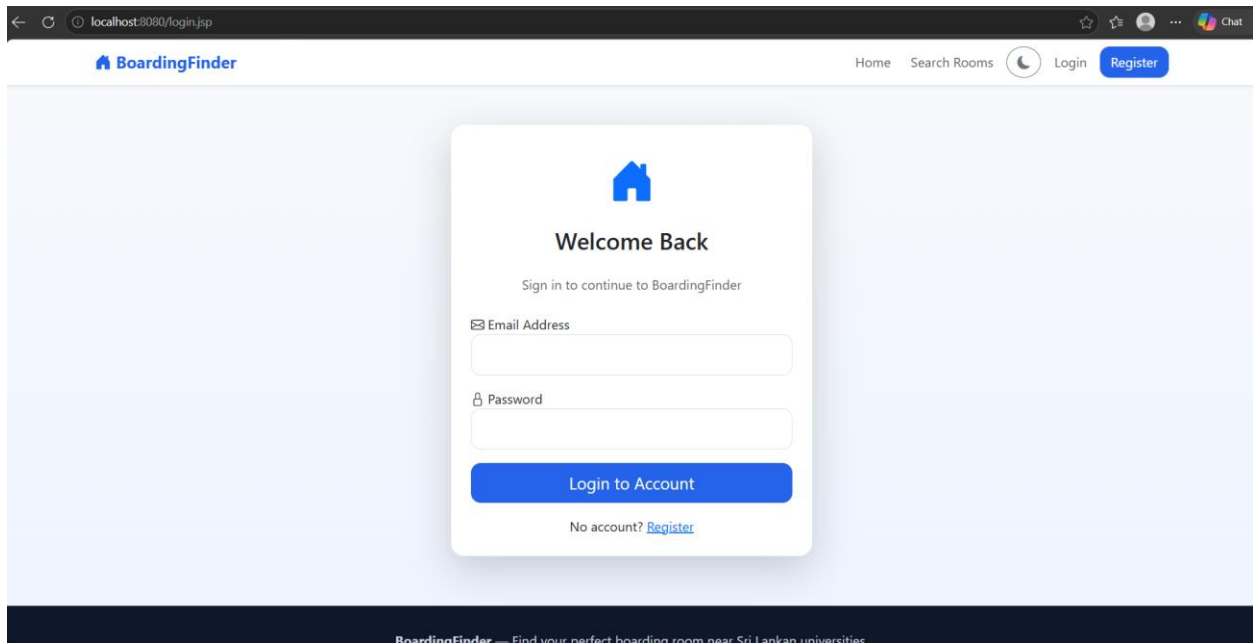
Register As

-- Select Role --

Create Account

Already have an account? [Sign In](#)

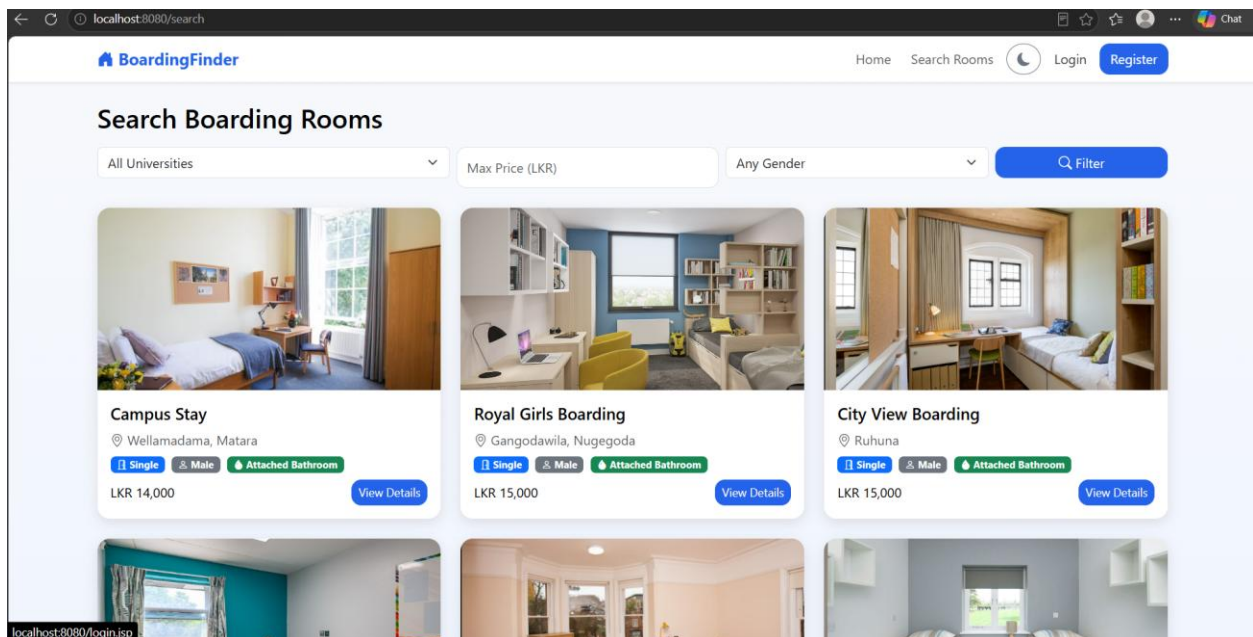
BoardingFinder — Find your perfect boarding room near Sri Lankan universities.
© 2026 BoardingFinder. All rights reserved.



5.3.2 Boarding Room Search

The search module enabled students to locate boarding rooms using filters such as university, gender preference, and maximum monthly rent. The search results displayed only available rooms matching the selected criteria, making it easier for students to identify suitable accommodation.

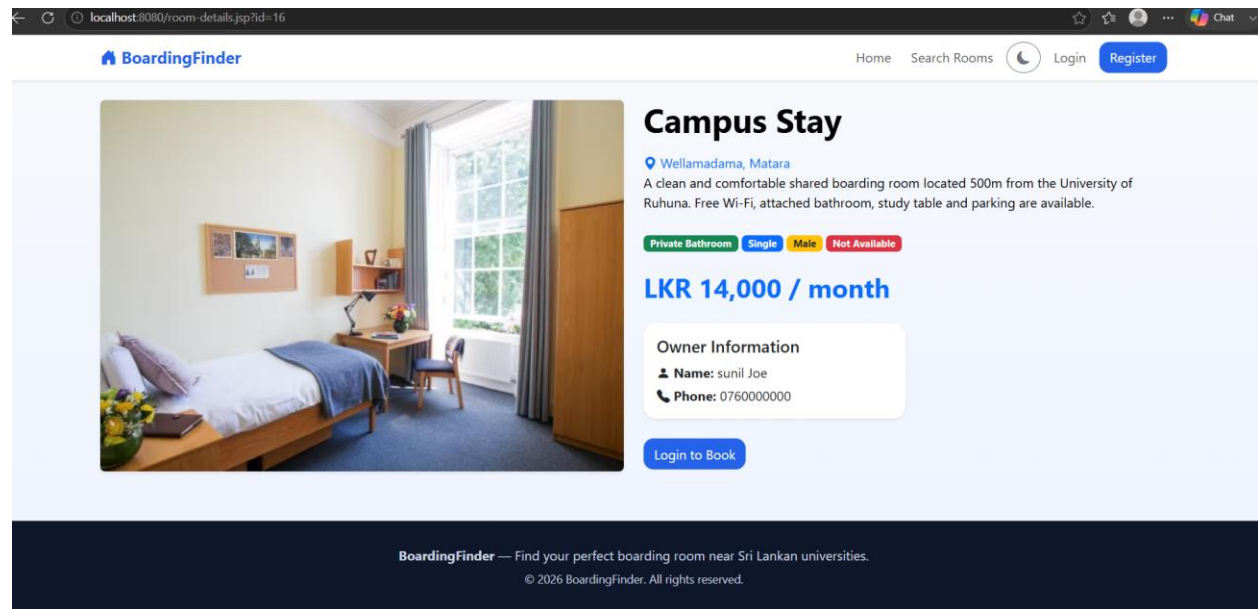
Figure 5.2: Boarding Room Search



5.3.3 Room Details

The room details page displayed comprehensive information about each boarding room, including photographs, rental fees, facilities, boarding location, Google Maps link, owner contact information, and room availability. This information helped students compare available options before submitting booking requests.

Figure 5.3: Room Details

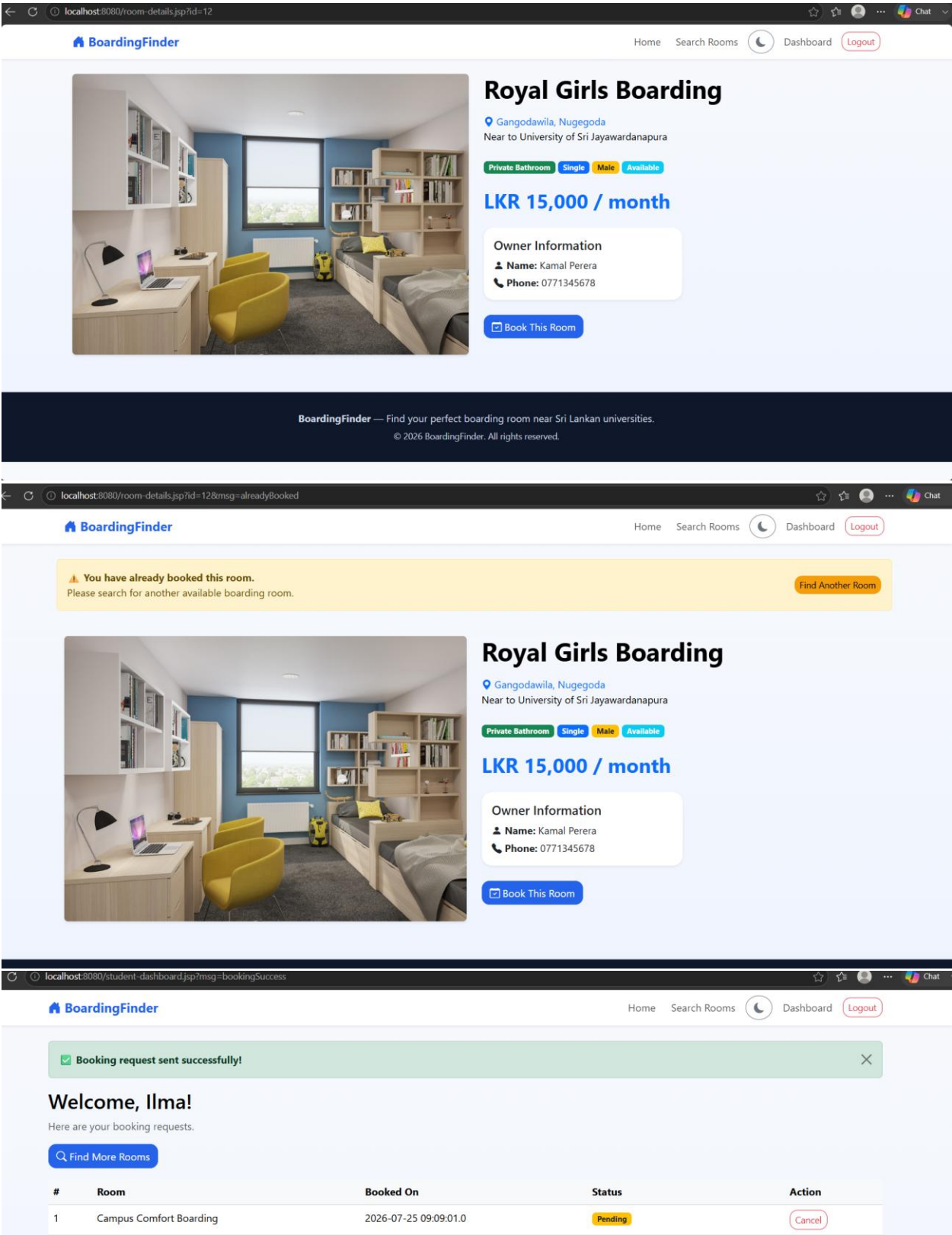


5.3.4 Booking Management

The booking module functioned successfully throughout testing. Students could submit booking requests for available rooms, while the system prevented duplicate booking requests. Newly submitted bookings were stored with a **Pending** status until reviewed by the boarding owner.

Boarding owners could accept or reject booking requests directly from the owner dashboard, and students were able to view the updated booking status in their dashboards.

Figure 5.4: Booking Management

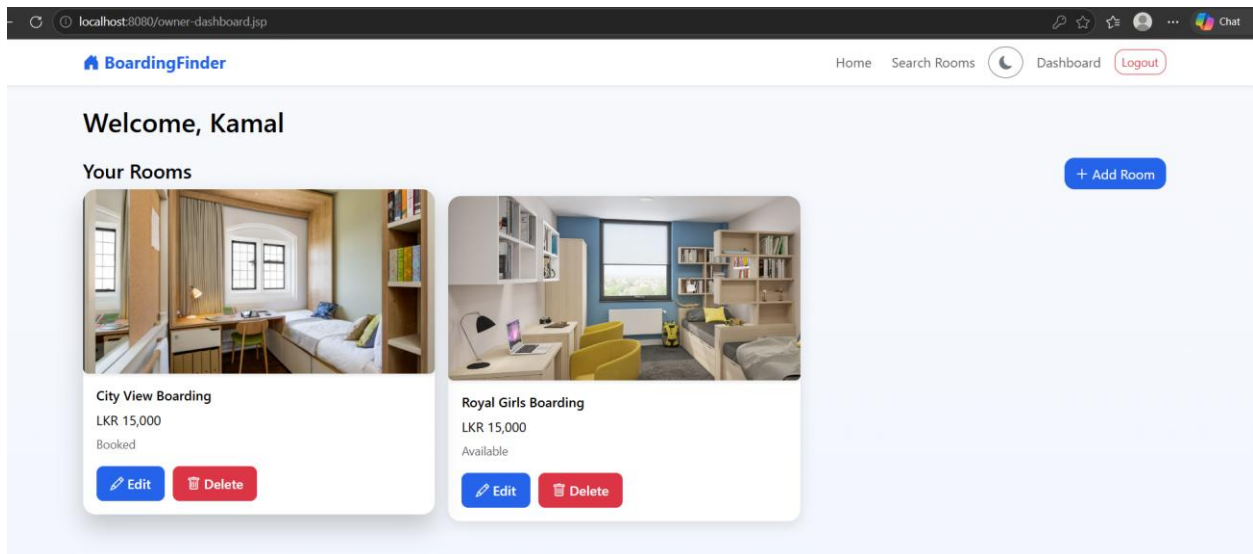


5.3.5 Room Management

Boarding owners successfully managed room information using the Add Room, Edit Room, and Delete Room functions. Changes made by the owner were immediately reflected in the room search results available to students.

The room management module simplified the maintenance of boarding information and reduced manual record-keeping.

Figure 5.5: Room Management



5.3.6 Image Upload

The image upload feature allowed boarding owners to upload photographs of their rooms. Uploaded images were displayed on both the search results page and the room details page, enabling students to evaluate boarding rooms visually before making a booking decision.

Figure 5.6: Image Upload

The screenshot shows the 'add-room.jsp' page of the BoardingFinder application. The form is used to add a new room listing. It includes fields for Title, Description, University (selected as Eastern University), Location, Room Type (selected as Single), Gender (selected as 1 Person), Maximum Occupants (selected as Any), Price (LKR), Attached Bathroom (checkbox), and Room Image (file upload area). The 'Save Room' and 'Cancel' buttons are at the bottom of the form. The top navigation bar includes links for Home, Search Rooms, Dashboard, and Logout, along with a Chat icon.

5.3.7 Automatic Capacity Management

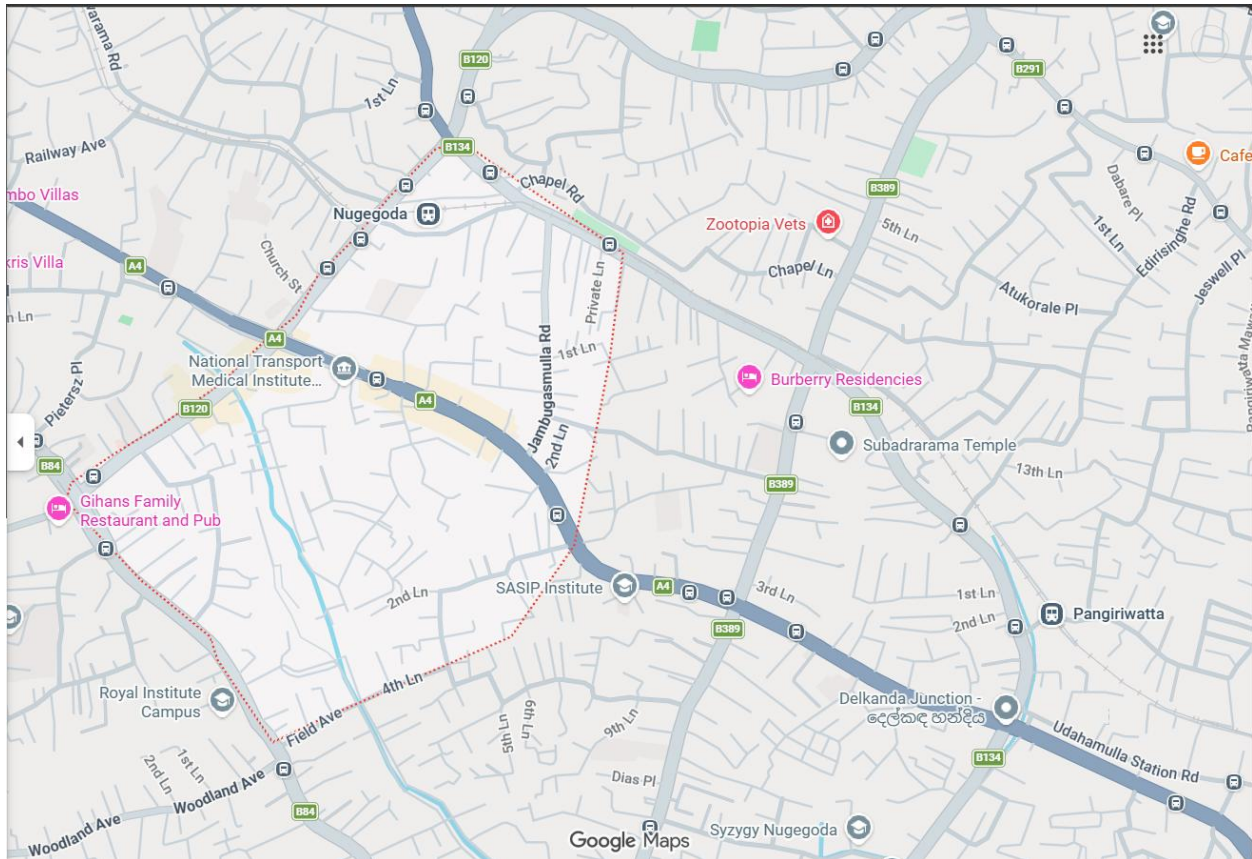
One of the major achievements of the system was the automatic management of room occupancy. Whenever a booking request was accepted, the system automatically increased the occupancy count. When the room reached its maximum capacity, its status changed to **Booked**, preventing additional bookings.

If an accepted booking was cancelled, the occupancy count decreased automatically, and the room became available again if capacity permitted. This functionality minimized manual work for boarding owners and ensured accurate room availability.

5.3.8 Google Maps Integration

The Google Maps feature enabled students to view the exact location of a boarding room before booking. This functionality made it easier to evaluate the distance between the boarding house and the university.

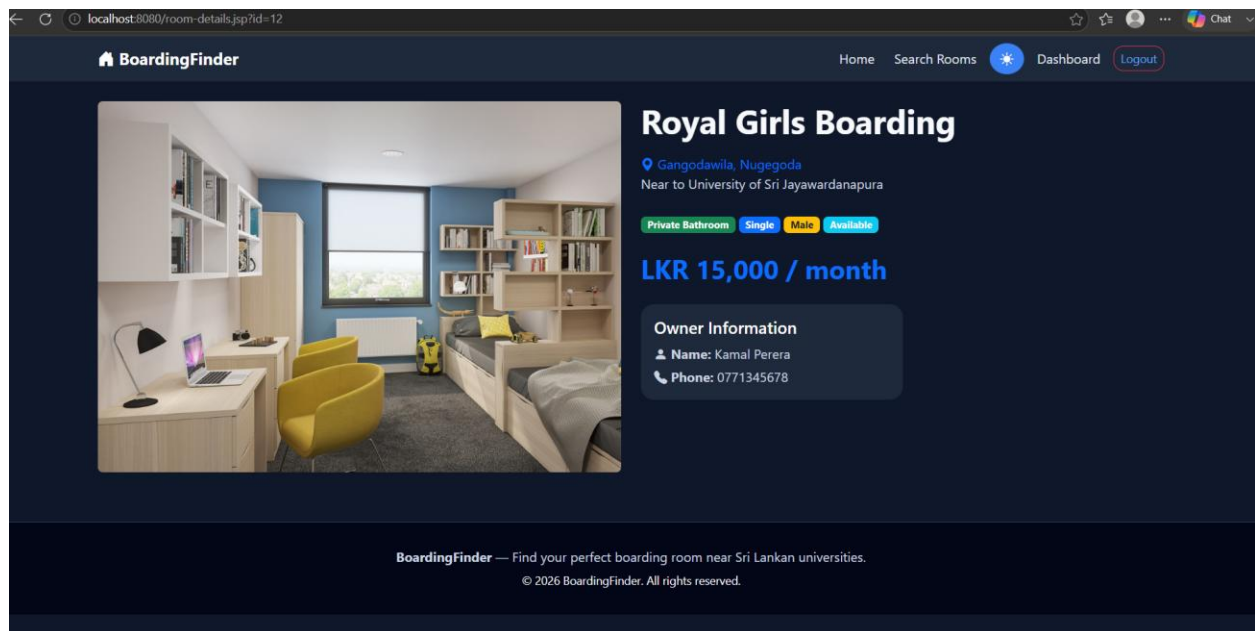
Figure 5.7: Google Maps Integration



5.3.9 Dark Mode and Light Mode

The system successfully provided both Light Mode and Dark Mode interfaces. Users could switch between themes based on their preferences. The feature enhanced user comfort, particularly during extended usage.

Figure 5.8: Dark Mode Interface



5.4 System Performance

The Boarding Finder System demonstrated satisfactory performance during testing. Page navigation was smooth, search results were displayed promptly, and database operations such as registration, login, room management, and booking processing were completed without significant delays.

The use of Java Servlets and JDBC enabled efficient communication with the MySQL database, while Bootstrap ensured that the user interface remained responsive on different screen sizes.

5.5 Advantages of the Developed System

The developed system provides several advantages over traditional manual boarding management methods.

- Centralized management of boarding information.
- Faster room searching using multiple filters.
- Easy booking process for students.

- Separate dashboards for students and boarding owners.
- Automatic room occupancy management.
- Prevention of duplicate booking requests.
- Google Maps integration for location viewing.
- Image upload support for boarding rooms.
- Responsive and user-friendly interface.
- Support for both Dark Mode and Light Mode.

5.6 Limitations

Although the Boarding Finder System achieved its intended objectives, several limitations remain.

- Online payment functionality has not been implemented.
- Email or SMS notifications are not available.
- User ratings and reviews are not included.
- Chat functionality between students and boarding owners is unavailable.
- The application currently supports only web browsers and does not include a mobile application.

These limitations provide opportunities for future enhancement.

5.7 Chapter Summary

This chapter presented the results obtained from implementing the Boarding Finder System and discussed the performance of each major module. The implemented system successfully satisfied the project objectives by providing an efficient platform for searching, booking, and managing boarding rooms. Testing demonstrated that the application performed reliably, while additional features such as automatic occupancy management, duplicate booking prevention, Google Maps integration, and theme switching improved the overall user experience.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Introduction

This chapter presents the overall conclusion of the Boarding Finder System and evaluates whether the project objectives were successfully achieved. It also discusses the limitations of the current system and suggests future enhancements that could improve the application's functionality and user experience.

6.2 Conclusion

The Boarding Finder System was successfully designed, developed, and implemented as a web-based application to simplify the process of searching and managing boarding accommodations for university students. The system addressed the problems associated with traditional methods of finding boarding places, which often involve manual communication, limited information, and time-consuming searches.

The developed application provides a centralized platform where students can search for boarding rooms using filters such as university, gender preference, and maximum monthly rent. Students can view detailed room information, including images, facilities, rental fees, Google Maps locations, and owner contact details before submitting booking requests.

The system also provides a dedicated dashboard for boarding owners, allowing them to add, edit, and delete room information, upload room images, and manage booking requests efficiently. The automatic occupancy management feature ensures that room availability is updated whenever bookings are accepted or cancelled. Furthermore, the system prevents duplicate booking requests and automatically changes the room status to **Booked** when the maximum capacity is reached.

The application was implemented using Java, Java Server Pages (JSP), Java Servlets, MySQL, HTML, CSS, JavaScript, Bootstrap, and JDBC. These technologies provided a reliable and scalable platform for developing the system.

System testing demonstrated that all major modules functioned correctly, including user registration, authentication, room searching, booking management, room management, image uploading, Google Maps integration, and theme switching. The application met the objectives defined during the planning stage and provided an efficient, user-friendly solution for both students and boarding owners.

Overall, the Boarding Finder System successfully improves the boarding room searching and management process by reducing manual work, increasing accessibility, and providing accurate and up-to-date boarding information.

6.3 Achievement of Project Objectives

The objectives defined at the beginning of the project were successfully achieved.

Objective	Achievement
Develop a web-based boarding management system	Achieved
Provide secure user registration and login	Achieved
Enable students to search boarding rooms	Achieved
Allow online booking requests	Achieved
Provide room management for owners	Achieved
Upload boarding room images	Achieved
Integrate Google Maps location	Achieved
Prevent duplicate bookings	Achieved
Automatically manage room occupancy	Achieved

Table 6.1: Achievement of Project Objectives

6.4 Limitations

Although the Boarding Finder System successfully meets its objectives, several limitations remain.

- The system does not support online payment processing.
- Email notifications are not available.
- SMS notifications are not implemented.
- Student reviews and rating functionality are unavailable.
- Live chat between students and boarding owners is not supported.
- The application is currently available only as a web application and does not include a mobile version.
- Advanced recommendation features based on user preferences are not included.

Despite these limitations, the system provides all the essential functionalities required for effective boarding room management.

6.5 Future Enhancements

Several improvements can be incorporated into future versions of the Boarding Finder System to enhance its usability and functionality.

Online Payment Integration

An online payment gateway can be integrated to allow students to pay booking fees or monthly rent securely through the application.

Email and SMS Notifications

Automatic notifications can be sent to students and boarding owners whenever booking requests are submitted, accepted, rejected, or cancelled.

Student Ratings and Reviews

Students can be allowed to rate boarding rooms and write reviews after their stay. This feature would help future students make informed decisions.

Real-Time Chat

A messaging feature can be implemented to enable direct communication between students and boarding owners.

Mobile Application

An Android and iOS version of the Boarding Finder System can be developed to improve accessibility and convenience.

Advanced Search

Future versions can provide additional search filters such as:

- Distance from university
- Room type
- Internet availability
- Parking facilities
- Meals included
- Pet-friendly accommodation

Recommendation System

Artificial Intelligence or machine learning techniques can be used to recommend suitable boarding rooms based on user preferences and previous searches.

Online Document Verification

Students may upload identity documents while owners verify tenant information before confirming bookings.

6.6 Final Conclusion

The Boarding Finder System successfully demonstrates how modern web technologies can be used to improve the boarding room management process. The system provides an efficient platform for students to search and reserve boarding rooms while enabling boarding owners to manage their properties through a centralized web application.

By incorporating features such as secure authentication, room searching, booking management, image uploading, Google Maps integration, automatic occupancy management, duplicate booking prevention, and responsive user interfaces, the project fulfills its intended objectives and provides a practical solution to real-world accommodation management challenges.

The system forms a strong foundation for future development and can be expanded with additional features such as online payments, mobile applications, notification services, and intelligent recommendation systems.

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Use the citation style required by your university (APA, IEEE, etc.). Below is an APA-style example:

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